



CHAPTER 2

ATOMIC STRUCTURE



2.1

Bohr's

Atomic Models

BOHR'S ATOMIC MODELS/THEORY

How the theory begins? From the Rutherford's model

01

Rutherford's atom was a big step forward in evolution of atomic theory.

02

BUT he does not indicate how electron are arranged outside the nucleus of an atom.

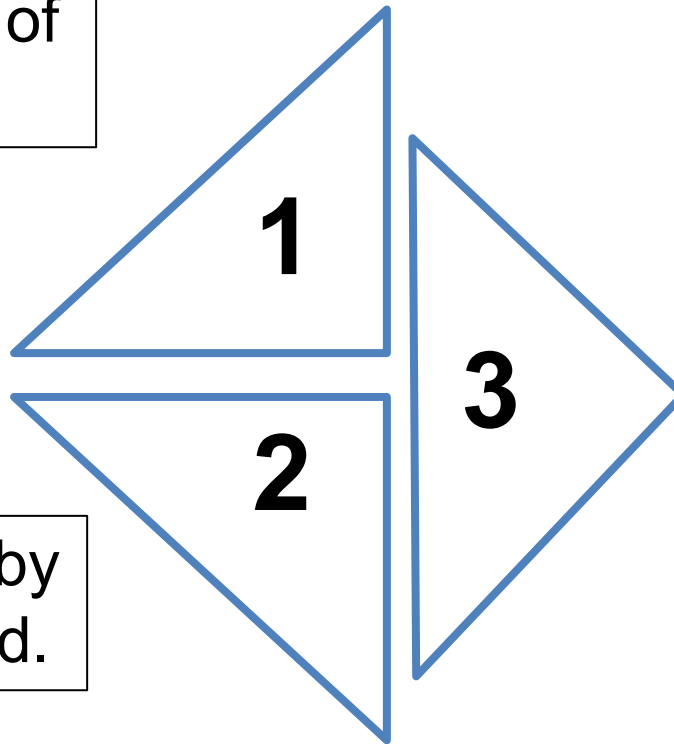
03

Then, Niels Bohr suggests that electron in an atom must be in motion and exist in specific region at various distance.

BOHR'S ATOMIC POSTULATES

Electron moves in circular path around the nucleus of an atom.

The energies possessed by the electron are quantised.



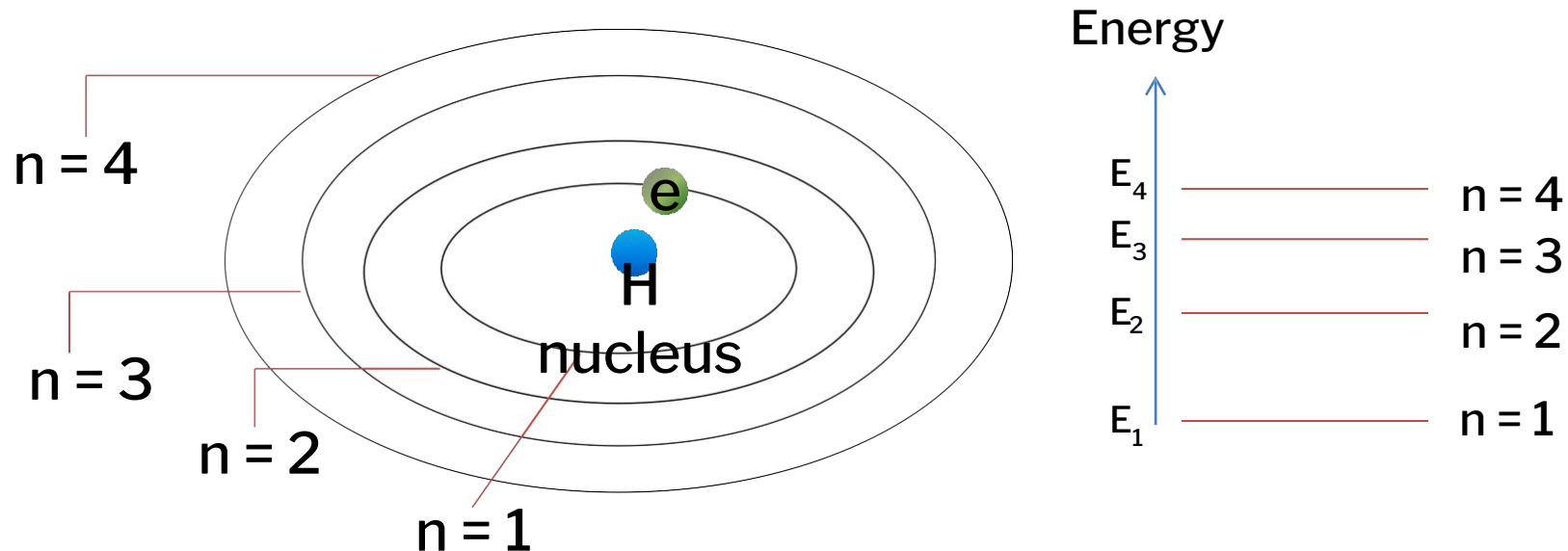
Energy is absorbed or emitted by an electron when it changes from one energy level to another. This energy is absorbed or emitted in the form of photons.

POSTULATE 1

- **Electron moves in circular path around the nucleus of an atom.**

When moving in the orbit, the electron **does not radiate or absorb any energy.**

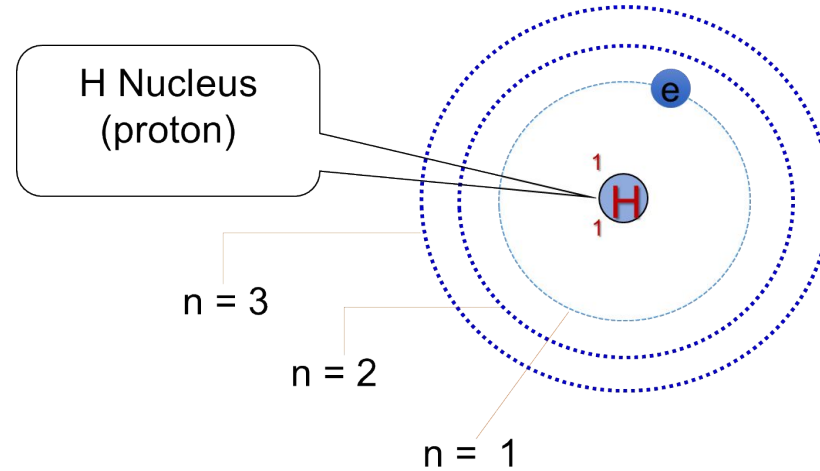
- The energy of the electron in an orbit is proportional to its distance from the nucleus.
- The farther the electron is from the nucleus, the more energy it has.
- Each orbit represent by an integer known as **principal quantum number, n .**



POSTULATE 2

- The energies possessed by the electron are quantised.

An electron moving in an orbit can only have a **certain amount of energy**.



The energies that the electron can possess at **specified energy level** are given by :


$$E_n = -R_H \left(\frac{1}{n^2} \right)$$

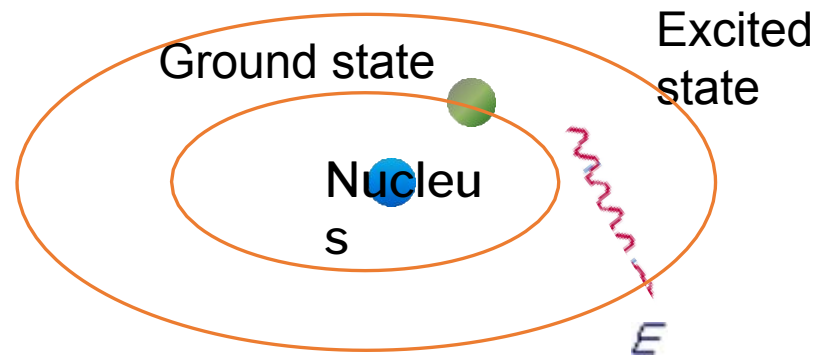
where

R_H = Rydberg constant, 2.18×10^{-18} J.

$n = 1, 2, 3, \dots, \infty$ (principal quantum number)

POSTULATE 3

- Energy is absorbed or emitted by an electron when it changes from one energy level to another. This energy is absorbed or emitted in the form of photons.
- ❖ The photon of energy **equals the energy difference between the levels.**



The **energy difference** between the **higher energy level (excited state)** and **lower energy level (ground state)** is given by:

$$\Delta E = E_f - E_i$$

$$\Delta E = -R_H \left(\frac{1}{n_f^2} \right) - \left(-R_H \left(\frac{1}{n_i^2} \right) \right)$$

Thus,

$$\Delta E = R_H \left(\frac{1}{n_i^2} - \frac{1}{n_f^2} \right)$$

$$R_H = -2.18 \times 10^{-18} \text{ J}$$

where :

Energy of an electron

$$E_i = -R_H \left(\frac{1}{n_i^2} \right)$$

$$E_f = -R_H \left(\frac{1}{n_f^2} \right)$$



- **Energy is emitted** by the electron when it falls from higher to lower energy level.
- The energy emitted is in the form of **photons**
 - The photon of energy produces particular **frequency (ν)** and **wavelength (λ)**

which is given by this:



$$\Delta E = h\nu$$

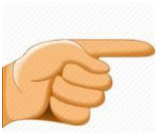
h (Planck's constant) = 6.6256×10^{-34} Js
 where:
 ν = frequency



$$\nu = \frac{c}{\lambda}$$

where c (the speed of light) = 3.00×10^8 ms⁻¹
 λ = wavelength

- Thus,



$$\Delta E = \frac{hc}{\lambda}$$

THE RYDBERG EQUATION

The **wavelength** corresponded to the transition of electron between two energy levels is calculated by using the Rydberg Equation.



$$\frac{1}{\lambda} = R_H \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \quad n_1 < n_2$$

where R_H

R_H = Rydberg constant, $1.097 \times 10^7 \text{ m}^{-1}$

$n = 1, 2, 3, \dots, \infty$ (principal quantum number)

Since λ should have a positive value thus $n_1 < n_2$

DIFFERENT VALUES OF R_H AND ITS USAGE.

1. $R_H = 1.097 \times 10^7 \text{ m}^{-1}$

$$\frac{1}{\lambda} = R_H \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \quad n_1 < n_2$$

2. $R_H = 2.18 \times 10^{-18} \text{ J}$

$$\Delta E = R_H \left(\frac{1}{n_i^2} - \frac{1}{n_f^2} \right)$$

H nucleus

Energy level

Energy associated with a specific orbit or state

Electron is excited.
A photon of energy
($\Delta E = +ve$)
is absorbed.

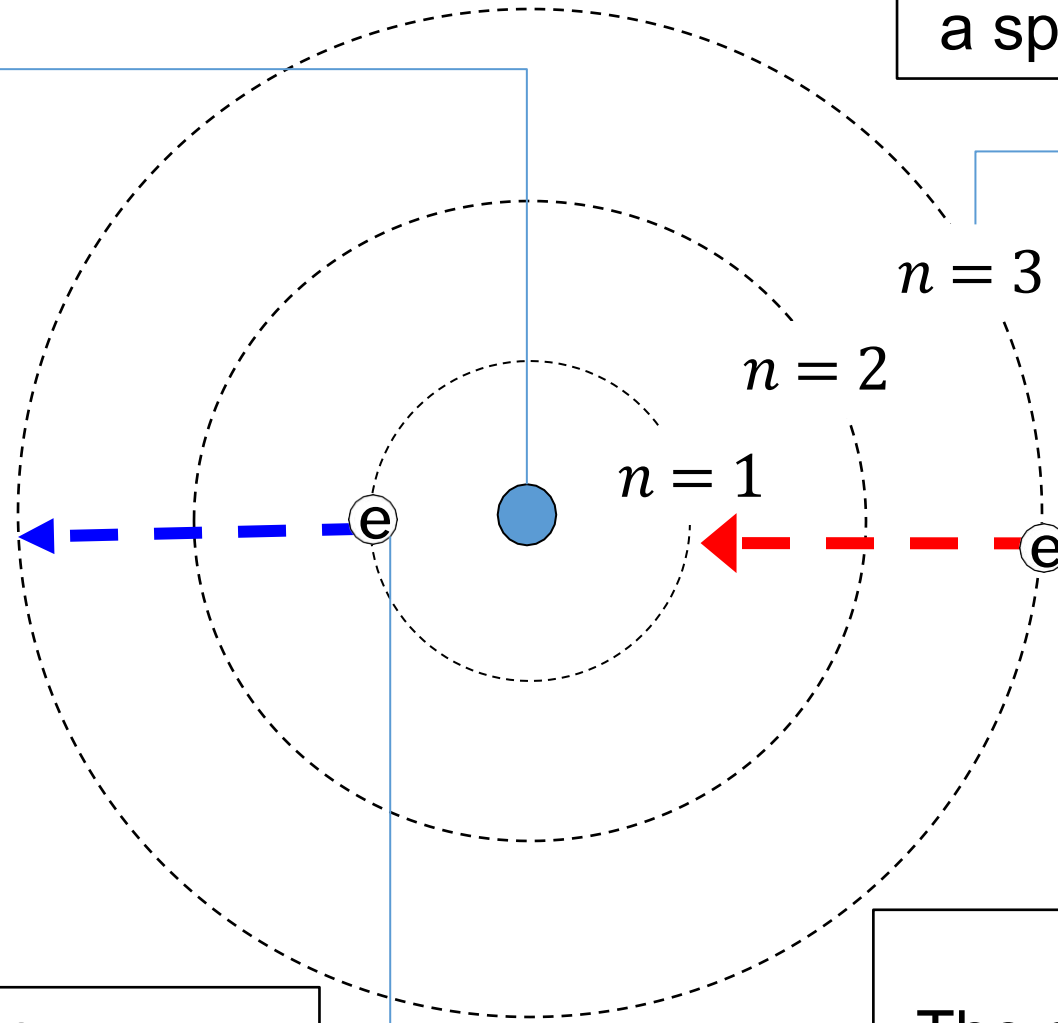
Electron falls back.
A photon of energy
($\Delta E = -ve$)
is released.

Ground state

The state in which the electrons
have their lowest energy

Excited state

The state in which the electrons
have shifted from lower energy
level to a higher energy level



Example 1

$$n = 2$$



Calculate the energy of an electron in the second orbit of a hydrogen atom?

Suggested answer:

$$E_n = -R \left(\frac{1}{n^2} \right)$$

$$= -2.18 \times 10^8 \text{ J} \left(\frac{1}{2^2} \right)$$

$$= -5.45 \times 10^{-19} \text{ J}$$

Example 2

- a) Calculate the energy released (J), that occurs when an electron falls from $n = 5$ to $n = 3$ energy level in a hydrogen atom.

$$\begin{aligned}\Delta E_R &= R_H \left(\frac{1}{n_i^2} - \frac{1}{n_f^2} \right) \\ &= 2.18 \times 10^{-18} \left(\frac{1}{5^2} - \frac{1}{3^2} \right) \\ &= 1.55 \times 10^{-19} \text{ J}\end{aligned}$$

Thus, energy released = $1.55 \times 10^{-19} \text{ J}$

REMARK:

The **negative sign** indicates that **energy is released** with an emission process.

- b) Calculate the frequency and wavelength (nm) of the radiation emitted.

$$\Delta E = h\nu$$

$$1.55 \times 10^{-19} \text{ J} = 6.6256 \times 10^{-34} \text{ Js} \times \nu$$

ν

$$\nu = 2.34 \times 10^{14} \text{ s}^{-1}$$

$$\nu = \frac{c}{\lambda}$$

$$\begin{aligned}&= \frac{3.00 \times 10^8 \text{ ms}^{-1}}{2.34 \times 10^{14} \text{ s}^{-1}} \\ &= 1.282 \times 10^{-6} \text{ m} = 1282 \text{ nm}\end{aligned}$$

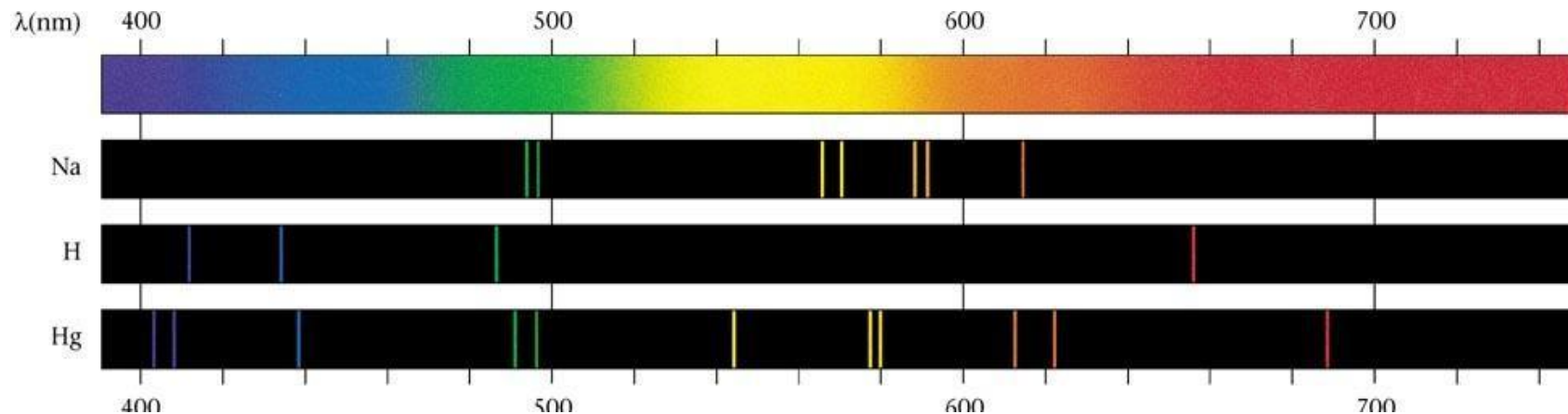
$$= 1.282 \times 10^{-6} \text{ m} = 1282 \text{ nm}$$

EMISSION SPECTRA

Emission Spectra

Continuous
Spectra

Line
Spectra



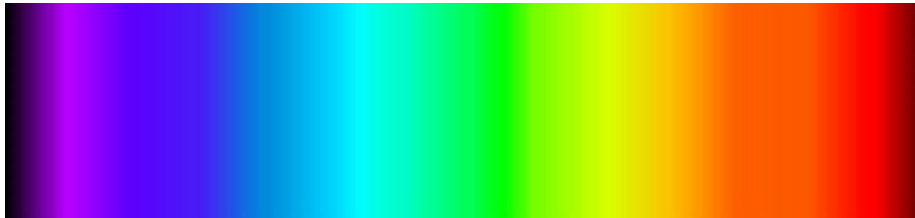
The comparison between continuous spectrum and line spectrum.

• Continuous Spectrum • Line Spectrum

A spectrum that consists of radiation of all wavelength.

Each color merges with no discrete or definite line that separate the colours.

Examples: Rainbow, sunlight, light from bulb



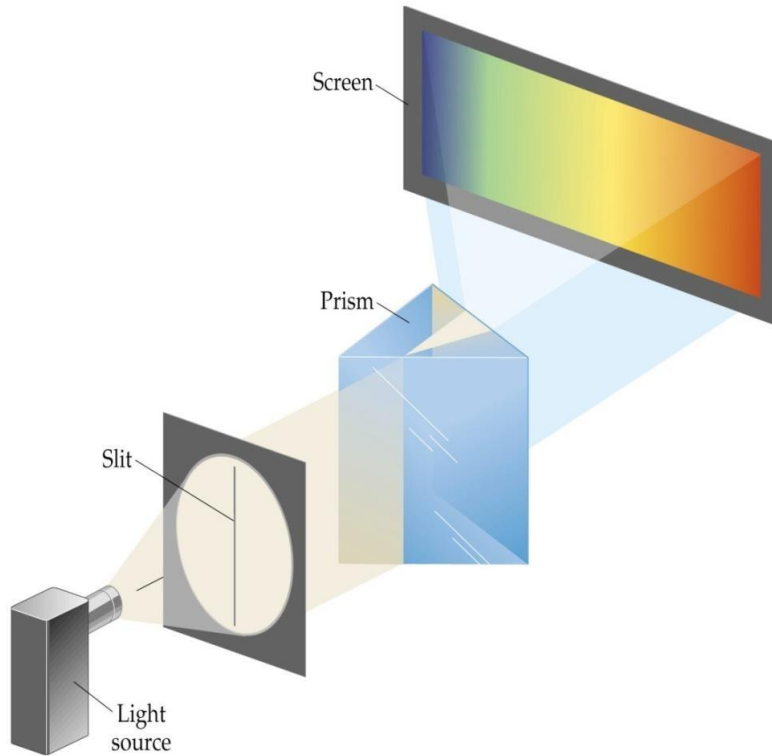
A spectrum that consists of radiation or series of discrete lines with specific wavelength.

Each line is discrete or definite which separated by blank areas

Example: emission spectrum of an atom @ element

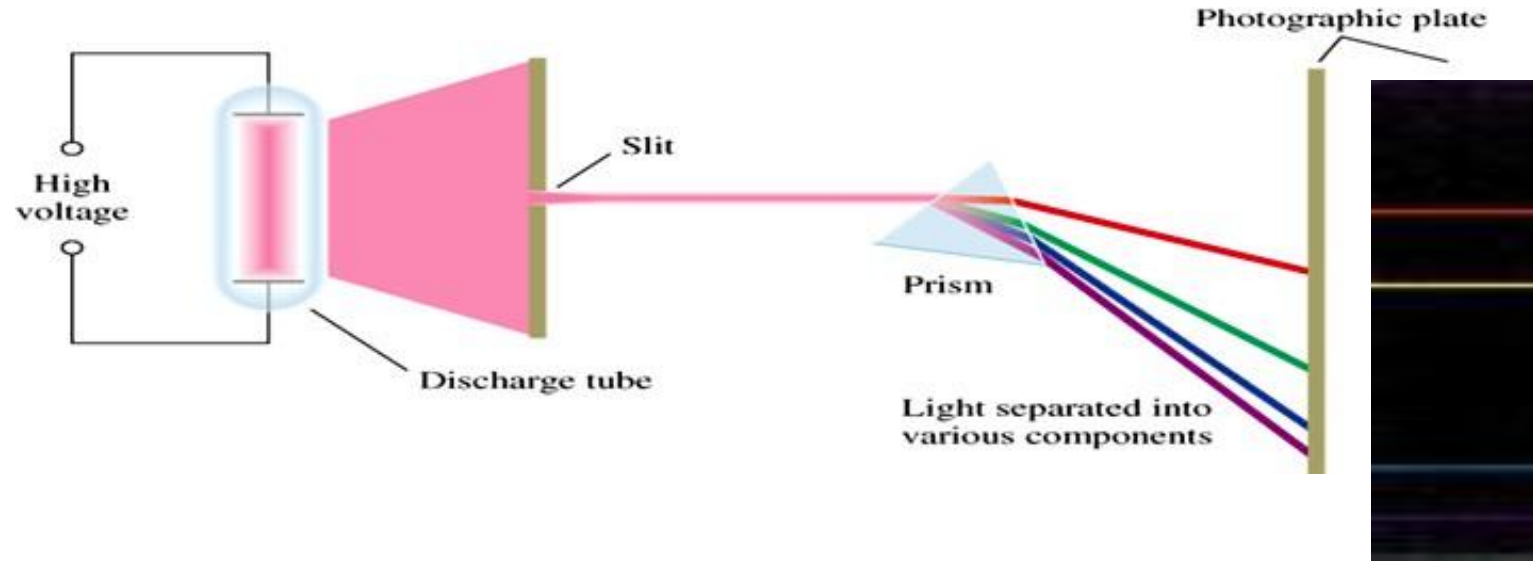


Formation of the Continuous Spectrum



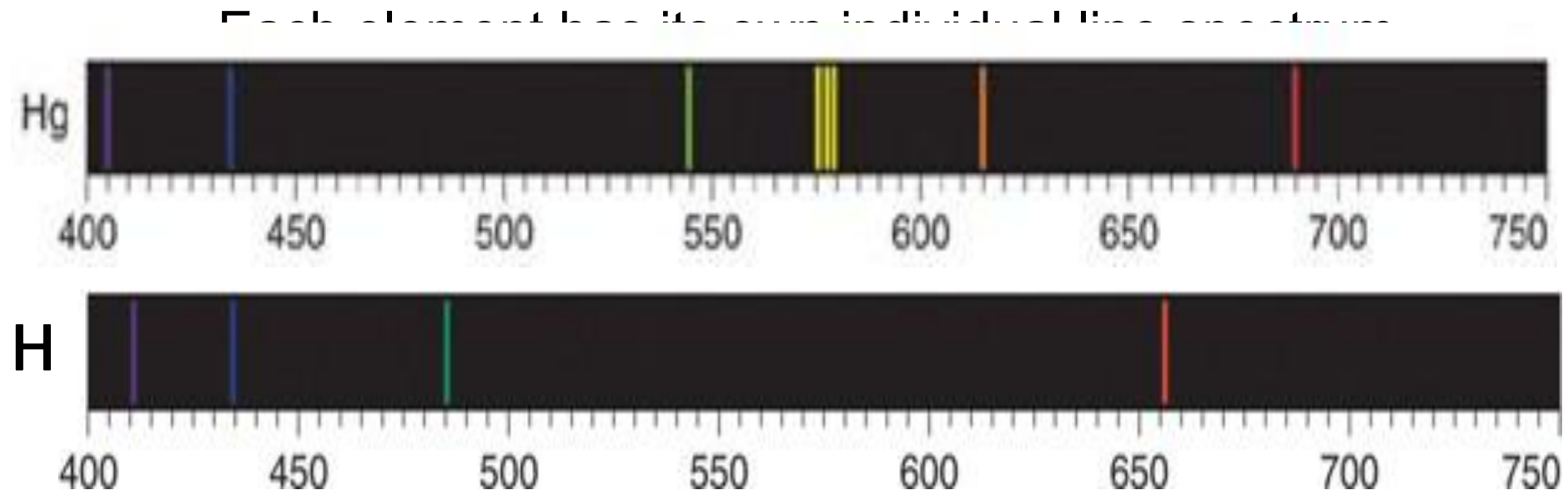
- When white light from luminous lamp is passed through a slit then a prism, it separates into a spectrum.
- The white light spread out into a rainbow of colours produces a continuous spectrum.
- The spectrum is **continuous** in that **all wavelengths are presents** and each colour merges into the next without a break.

LINE SPECTRUM

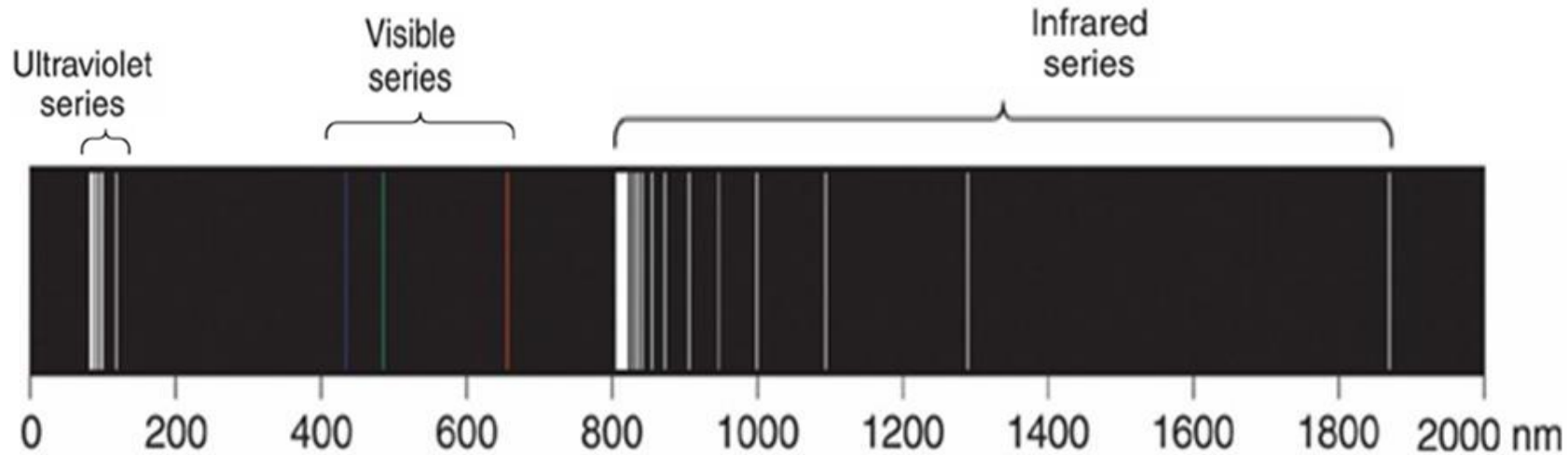


- If the source of light is from a discharge tube which contains a **specific element**, LINE SPECTRUM is produced.
- Line spectrum is a spectrum consists of **discontinuous and discrete lines** produced by excited atoms and ions as the electrons fall back to a lower energy level.
- The radiation emitted is **only at a specific wavelength or frequency**.

- It means **each line corresponds to a specific wavelength or frequency.**
- Line spectrum is composed of only a few wavelengths giving a series of discrete line separated by blank areas.

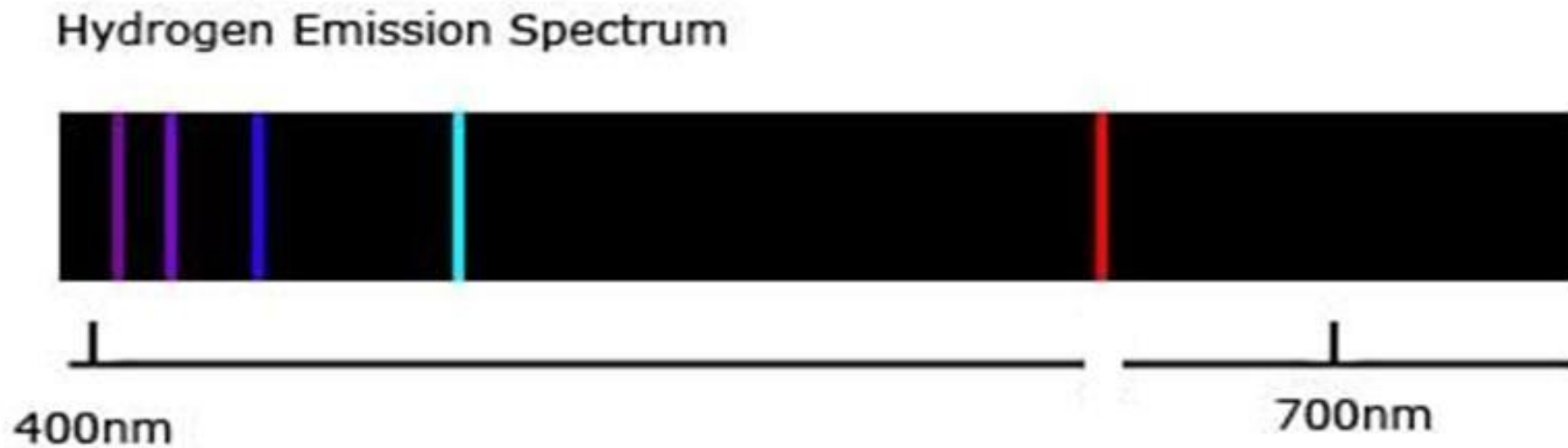


- What you would see is only a small part of the line spectrum.
- There is a lot more to the spectrum than the lines you can see with the naked eyes.
- Most of the spectrum is invisible to the eye because it is either in the infra-red or the ultra-violet.



- The spectrum is produced by the excited electrons as they **fall back from higher energy level to a lower energy level.**

With these conditions Bohr was able to explain the stability of atoms as well as the emission spectrum of hydrogen atom.



The emission spectrum of hydrogen is particularly important because most of the universe is made of hydrogen.

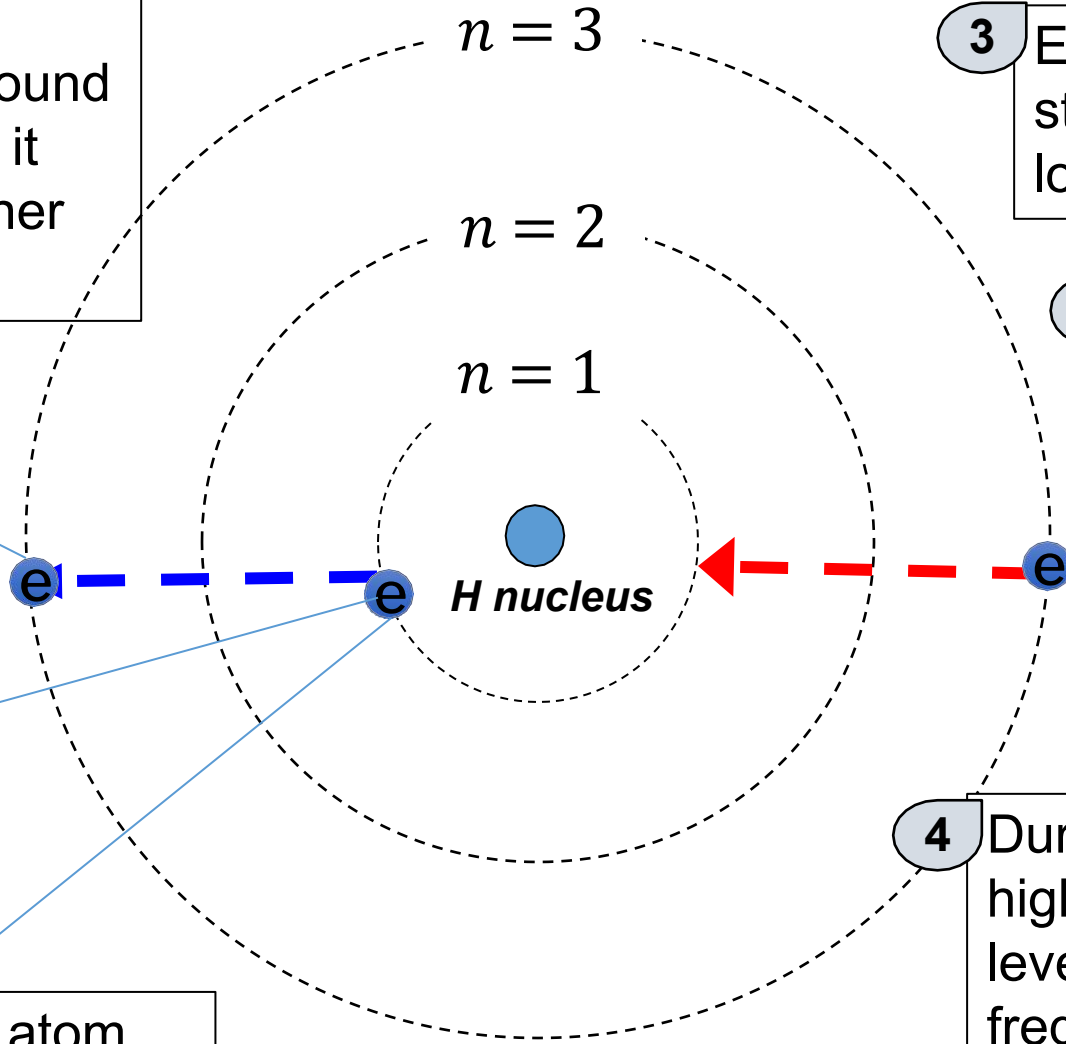
FORMATION OF HYDROGEN EMISSION SPECTRUM

2 When the electron of Hydrogen atom at its ground state, absorbs energy it is promoted to the higher energy level.

Electron at excited state

Electron at ground state

1 Electron in a hydrogen atom possesses a quantized of energy



3 Electron is unstable at the excited state and fall back from higher to lower energy level.

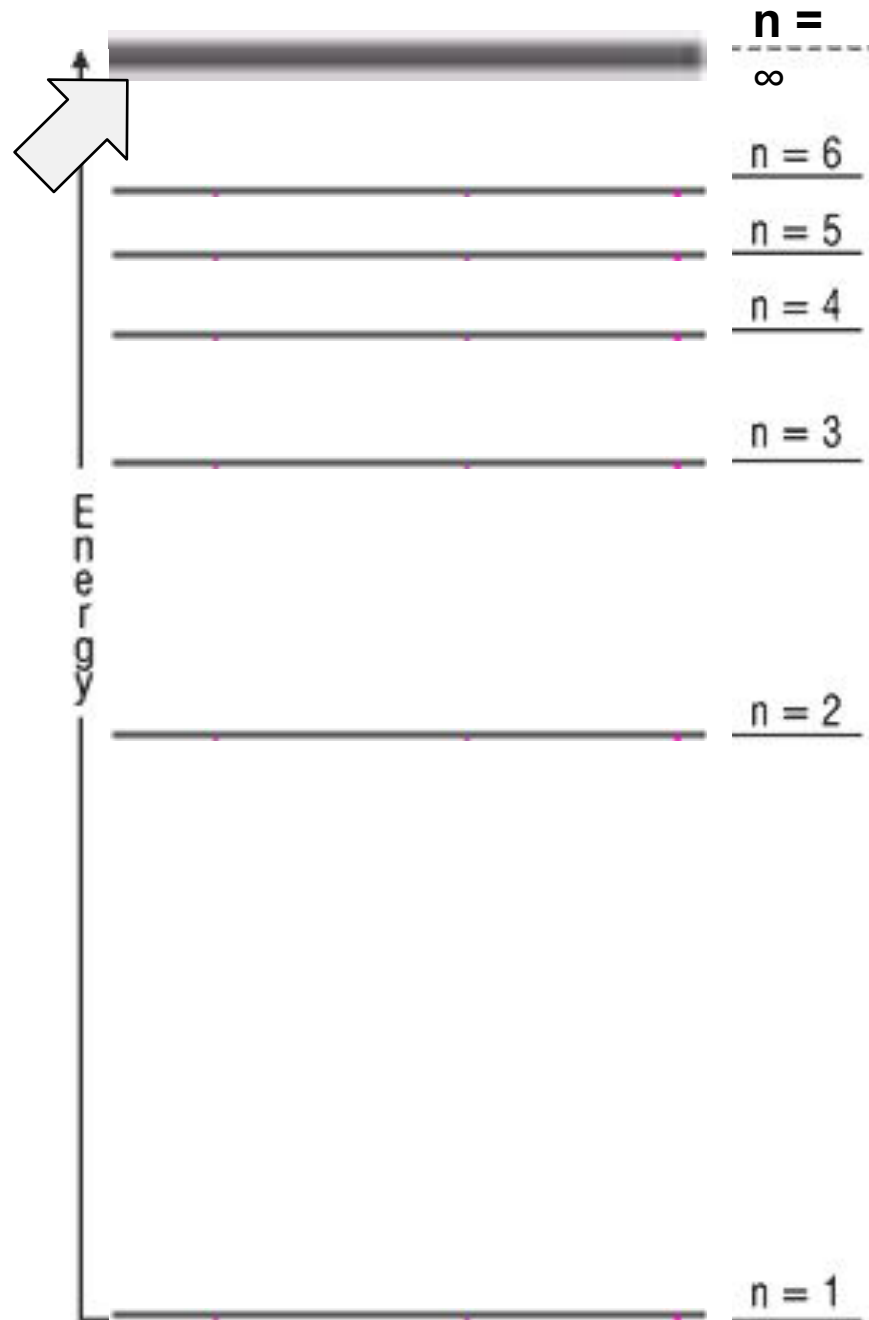
5 These photon (light) produce lines in the Hydrogen spectrum

4 During the transition of electron from higher energy level to lower energy level, the radiant energy with specific frequency is emitted in form of photon (light) which corresponding to a certain wavelength

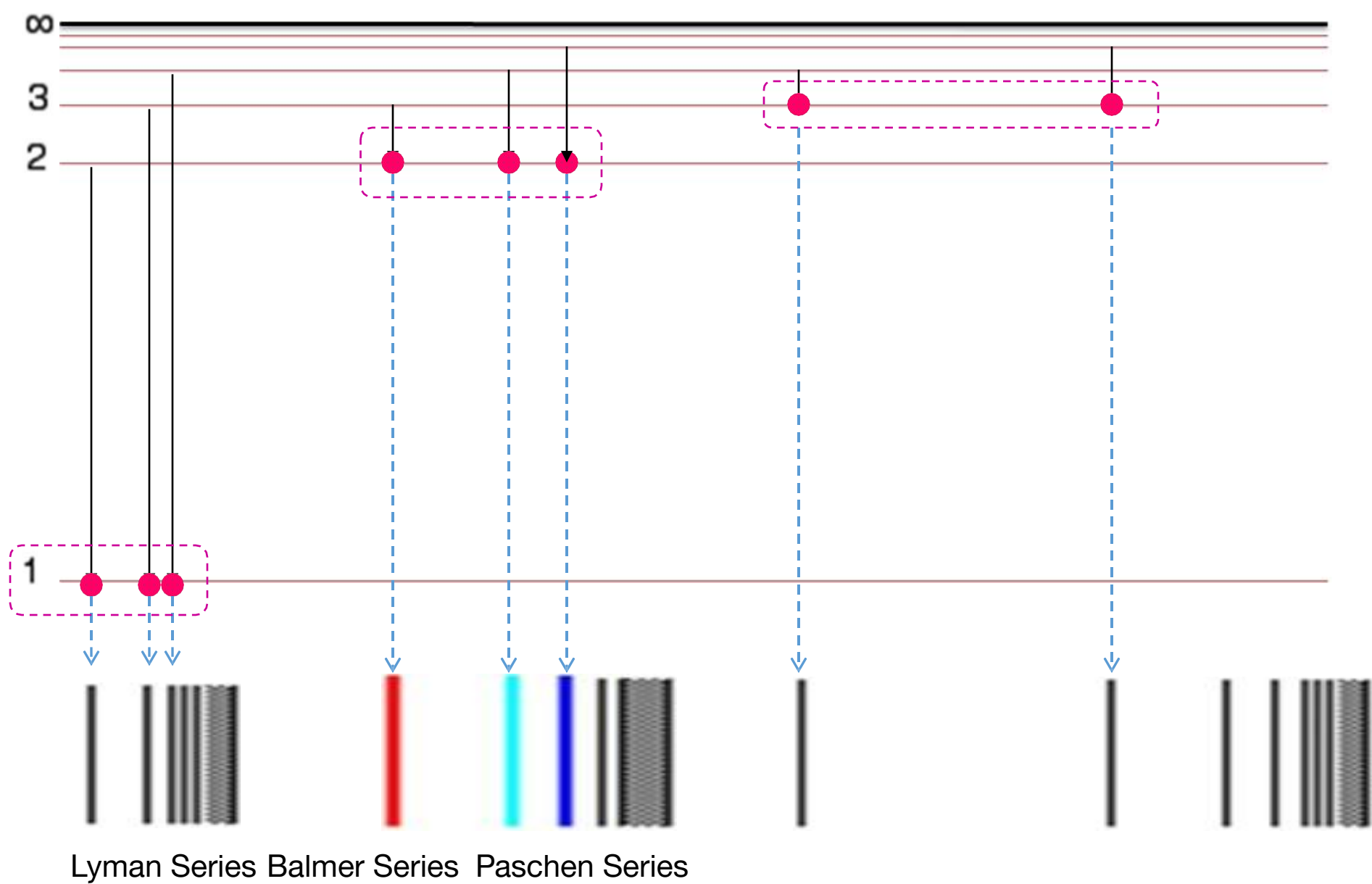
$$\Delta E = E_f - E_i = E_1 - E_3 (-ve)$$

Convergence limit

The wavelength or frequency at which the spectral lines merge together because the difference between energy levels become smaller and smaller.

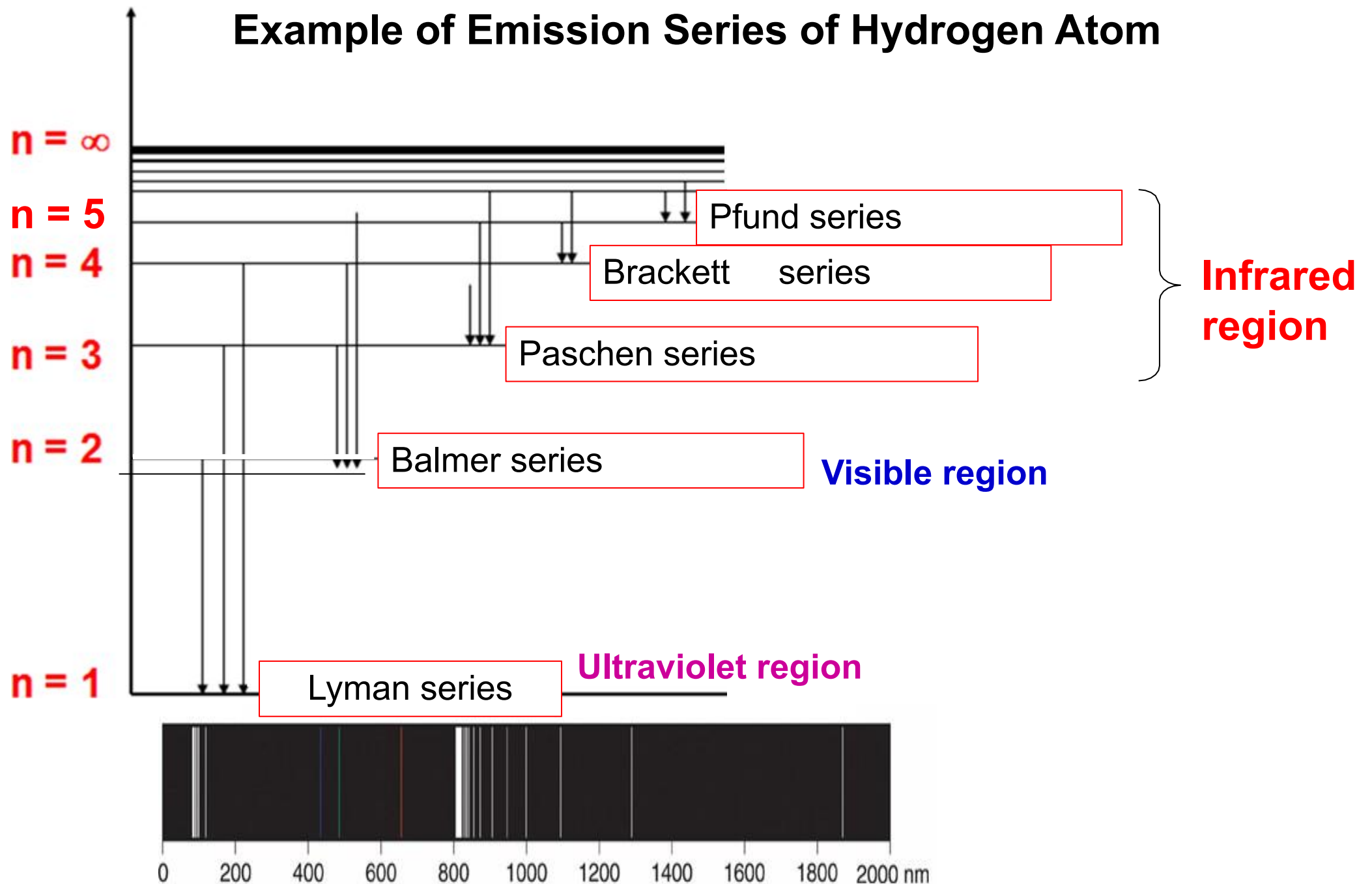


- The transition of electrons from the higher energy level to the lower energy level is shown by using an **energy level diagram**.
- The **intervals** between the energy level becomes **smaller and smaller** towards the top.
- Eventually, the lines become so close together that they practically form a continuous line, called **continuum limit @ convergence limit**.



- The groups of lines are **classified according to which level the electrons drop** and are **named after their discoverers**.

Example of Emission Series of Hydrogen Atom



Example 3

Complete the following table:

Series	n_1	n_2	Spectrum region
Lyman	1	2,3,4,...	Ultraviolet
Balmer	2	3,4,5,...	Visible
Paschen	3	4,5,6,...	Infrared
Brackett	4	5,6,7,...	Infrared
Pfund	5	6,7,8,...	Infrared

Example 4

Calculate the wavelength for a spectral line produced when an electron falls from $n = 5$ to $n = 3$. Name the series and state the region of electromagnetic spectrum for the line.

Suggested answer:

$$\frac{1}{\lambda} = R_H \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \quad n_1 < n_2$$

$$\frac{1}{\lambda} = 1.097 \times 10^7 \left(\frac{1}{3^2} - \frac{1}{5^2} \right)$$

$$\frac{1}{\lambda} = 7.80 \times$$

$$\frac{1}{\lambda} = 1.28 \times 10^{-6} \text{ m}$$

- Paschen Series
- Infrared region

Example 5

The first line of the Balmer series occurs at the wavelength of 656.3 nm. What is the energy difference between the two energy levels involved in the emission that results in this spectral line?

Suggested answer:

$$\begin{aligned}\Delta E &= \frac{hc}{\lambda} \\ &= \frac{(6.6256 \times 10^{-34} \text{ J}\cdot\text{s})(3 \times 10^8 \text{ m/s})}{656.3 \times 10^{-9} \text{ m}} \\ &= 3.03 \times 10^{-19} \text{ J}\end{aligned}$$

Thus, energy difference:

$$= -3.03 \times 10^{-19} \text{ J}$$

Alternative method:

$$\begin{aligned}\Delta E &= R_H \left(\frac{1}{n_i^2} - \frac{1}{n_f^2} \right) \\ &= 2.18 \times 10^{-18} \text{ J} \left(\frac{1}{3^2} - \frac{1}{2^2} \right) \\ &= -3.03 \times 10^{-19} \text{ J}\end{aligned}$$

REMARK:

The **negative sign** indicates that **energy is released** with an emission process.

Example 6

The Brackett series of the line spectrum of hydrogen is in the infrared region. The wavelength for the third line in the series is 2166 nm. Calculate the energy involved in the formation of the third line in Brackett series.

Suggested answer:

$$\begin{aligned}\Delta E &= \frac{hc}{\lambda} \\ &= \frac{(6.6256 \times 10^{-34}) (3 \times 10^8)}{2166 \times 10^{-9}} \\ &= 9.18 \times 10^{-20} \text{ J}\end{aligned}$$

Alternative method:

$$\begin{aligned}\Delta E &= R_H \left(\frac{1}{n_i^2} - \frac{1}{n_f^2} \right) \\ &= 2.18 \times 10^{-18} \text{ J} \left(\frac{1}{7^2} - \frac{1}{4^2} \right) \\ &= -9.18 \times 10^{-20} \text{ J}\end{aligned}$$

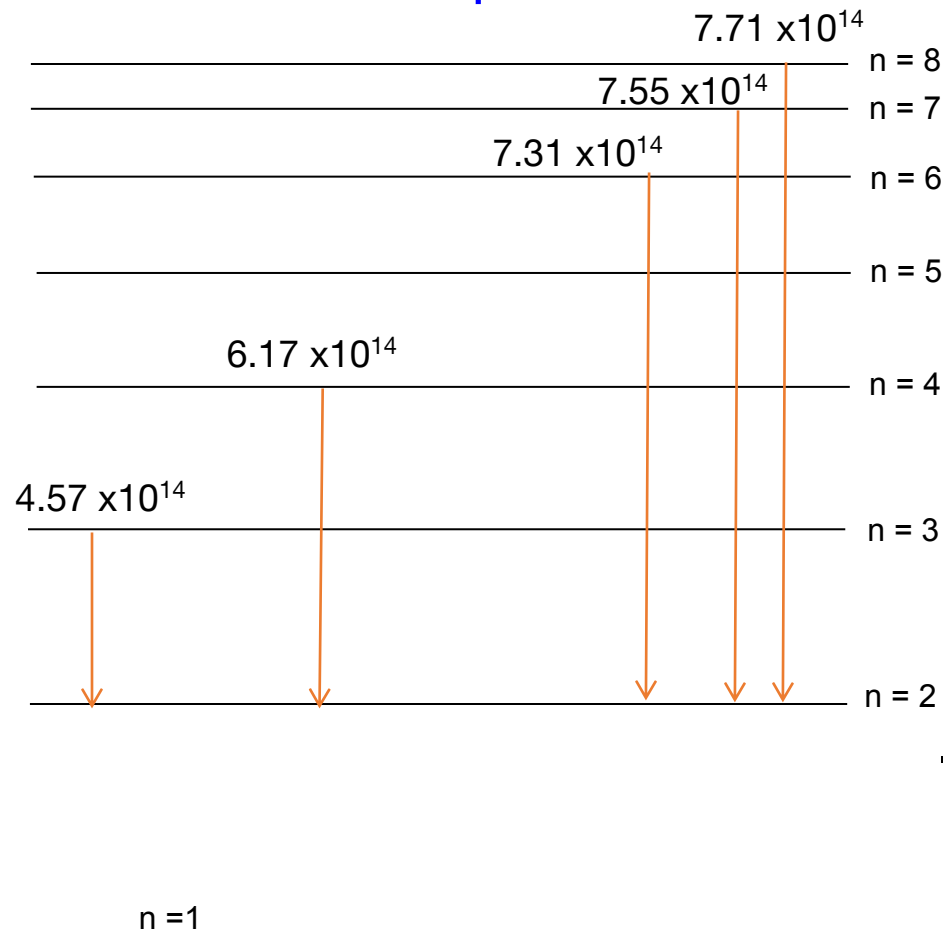
Thus, energy released $= 9.18 \times 10^{-20} \text{ J}$

Example 7

The frequencies of five lines in the Balmer series of the emission spectrum of hydrogen atom are as follows :

{ 4.57, 6.17, 7.31, 7.55 and 7.71 } ($\times 10^{14}$ Hz)

i) Copy the energy level diagram given below and show the electron transitions in hydrogen atom which correspond to **each of the frequency values**.



$$\Delta E = h\nu$$

$$= \{ 6.6256 \times 10^{-34} \} \{ 4.57 \times 10^{14} \}$$

$$= 3.03 \times 10^{-19}$$

J

$$= 3.03 \times 10^{-19}$$

$\therefore$ Energy emitted J

$$\Delta E = R_H \left(\frac{1}{n_i^2} - \frac{1}{n_f^2} \right)$$

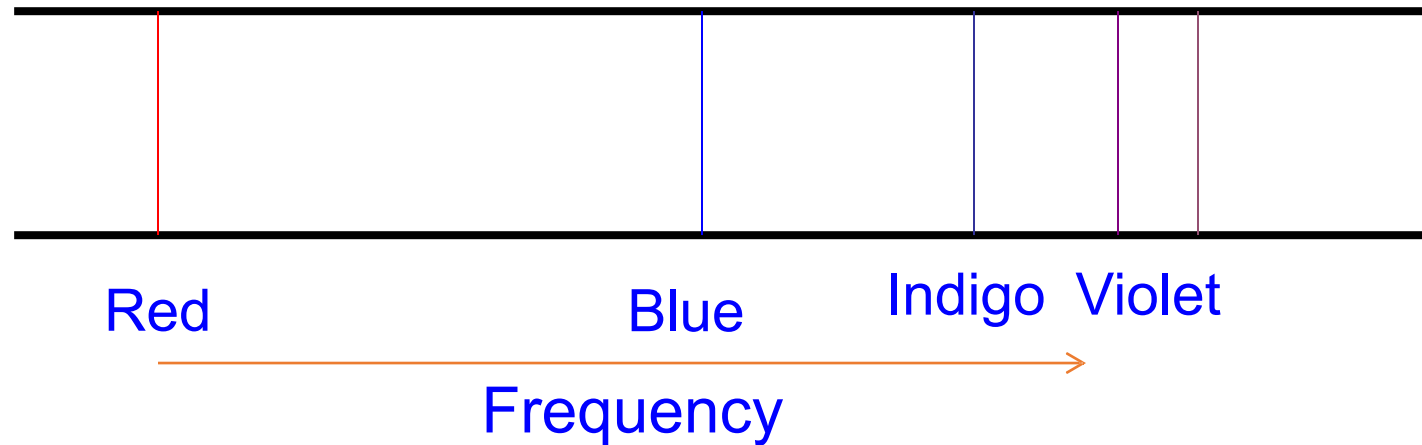
$$= 2.18 \times 10^{-18} \text{ J} \left(\frac{1}{2^2} - \frac{1}{n_i^2} \right)$$

$$n_i = 3$$

$$-3.03 \times 10^{-19}$$

Example 8

Figure shows the line spectrum of hydrogen atom in the visible region.



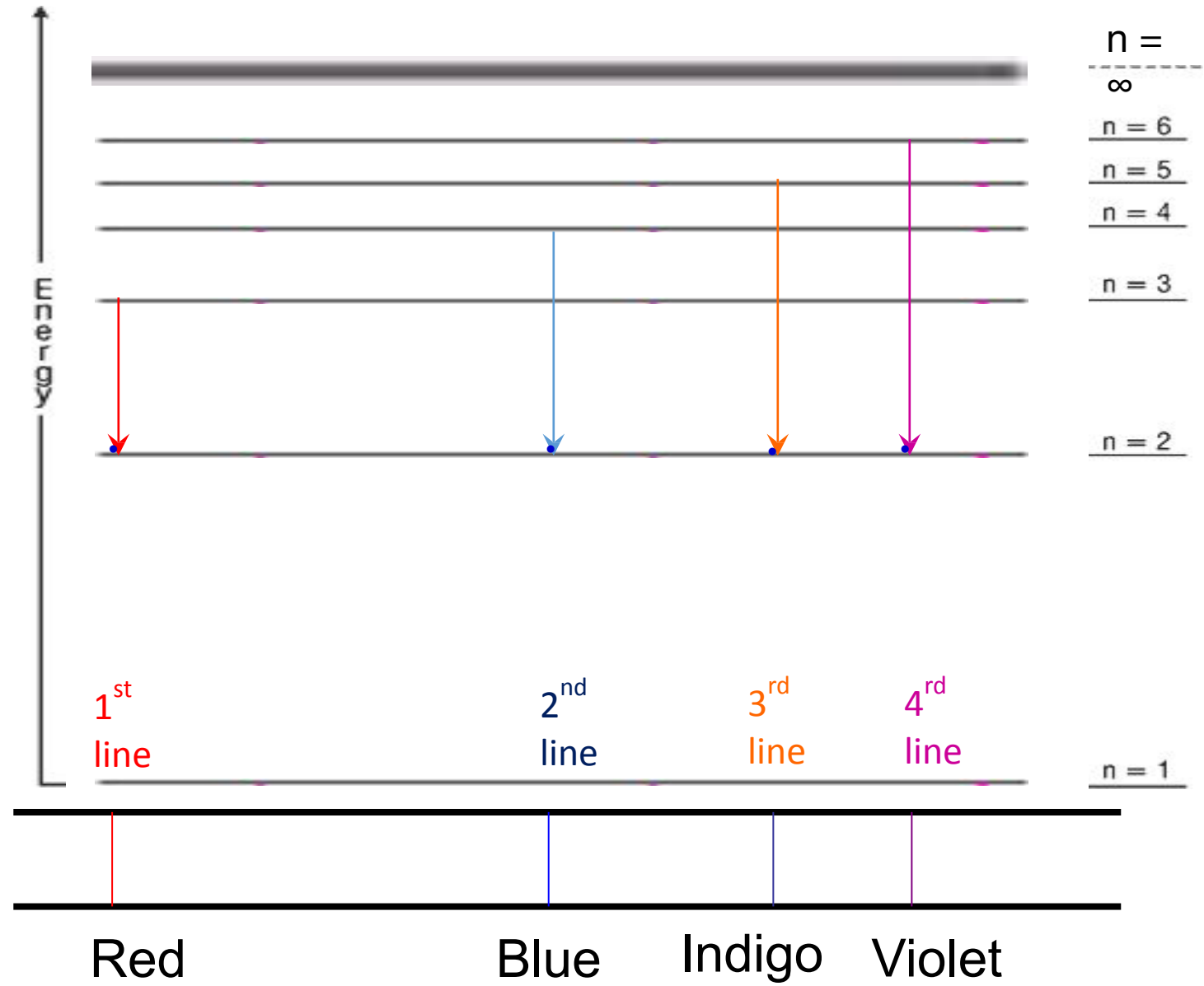
- Draw the energy levels in a hydrogen atom.
- Show the electron transitions which produce the above lines.
- Give the name of this particular series of lines.

[5 marks]

Suggested answer:

- i)
- ii)

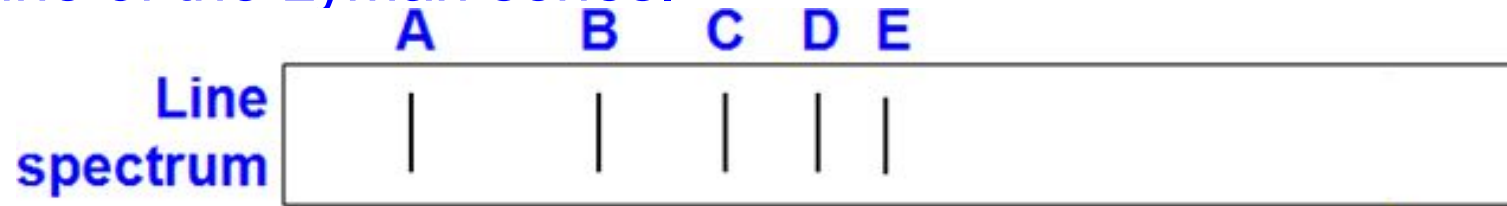
Visible region
(Ground state, $n = 2$)



iii) Balmer series

Example 9

The following diagram depicts the line spectrum of hydrogen atom. Line A is the first line of the Lyman series.



a) Specify the increasing order of the radiant energy, frequency and wavelength of the emitted photon.

Suggested answer:

□ Energy increase from line A to E.

$$\Delta E = h\nu; \quad E \propto \nu$$

$\therefore$ when $E \uparrow$, $\nu \uparrow$.

Thus, frequency increase from line A to E.

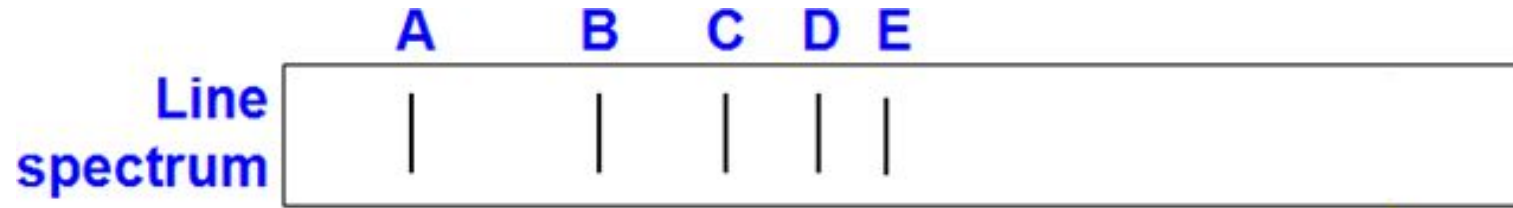
$$\nu = \frac{c}{\lambda}; \quad \frac{1}{\nu \lambda} \propto -$$

$\therefore$ when $\nu \uparrow$, λ
 $\downarrow$.

Thus, wavelength increase from line E to A.

Example 9 - continue

The following diagram depicts the line spectrum of hydrogen atom. Line A is the first line of the Lyman series.



- b) Which of the line that corresponds to
- i) the shortest wavelength?
 - ii) the lowest frequency?

Suggested answer:

- i) **Line E**
- ii) **Line A**

Example 10

State the transitions of electrons that lead to the lines W, and Y, respectively.



Suggested answer:

For W : transition of electron is from $n = 4$ to $n =$

2. For Y : electron shifts from $n = 7$ to $n = 2$.

Example 11

Calculate the wavelength of the third line in the Balmer series.

Suggested answer:

$$\Delta E = R_H \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

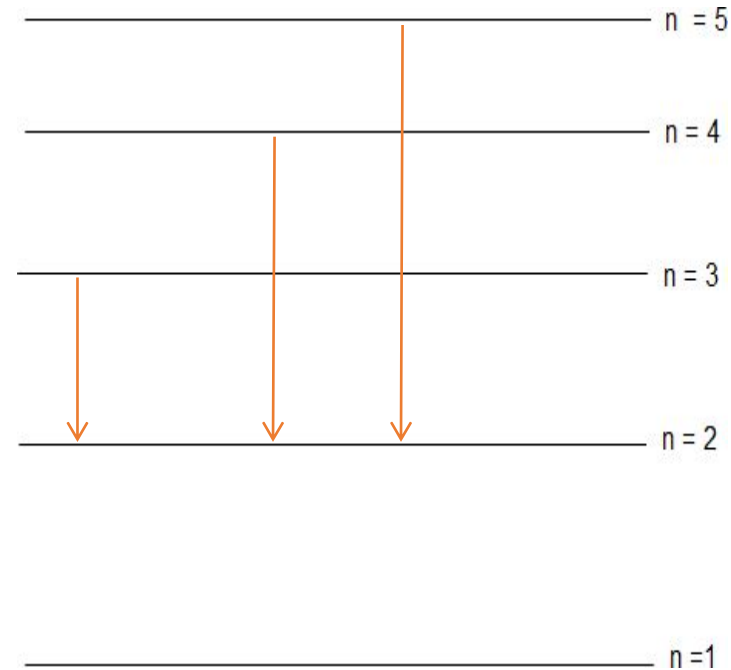
$$= 2.18 \times 10^{-18} \text{ J} \left(\frac{1}{2^2} - \frac{1}{5^2} \right)$$

$$= -4.58 \times 10^{-19} \text{ J}$$

$$\Delta E = \frac{hc}{\lambda}$$

$$4.58 \times 10^{-19} = \frac{(6.6256 \times 10^{-34} \text{ J}\cdot\text{s})(3 \times 10^8 \text{ m/s})}{\lambda}$$

$$\lambda = 4.34 \times 10^{-7} \text{ m @ } 434 \text{ nm}$$



REMARK:

To calculate the wavelength, omit the minus sign for ΔE because the wavelength of the photon must be positive.

Alternative method:

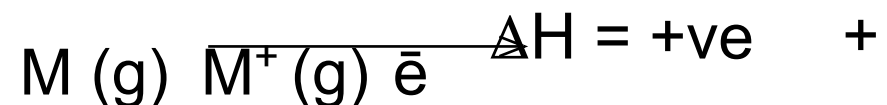
$$\frac{1}{R} = \frac{1}{4} \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \quad n_1 < n_2$$

$$\frac{1}{\lambda_{10}} = 1.097 \times \left(\frac{1}{2^2} - \frac{1}{5^2} \right)$$

$$\lambda = 4.34 \times 10^{-7} \text{ m @ 434 nm}$$

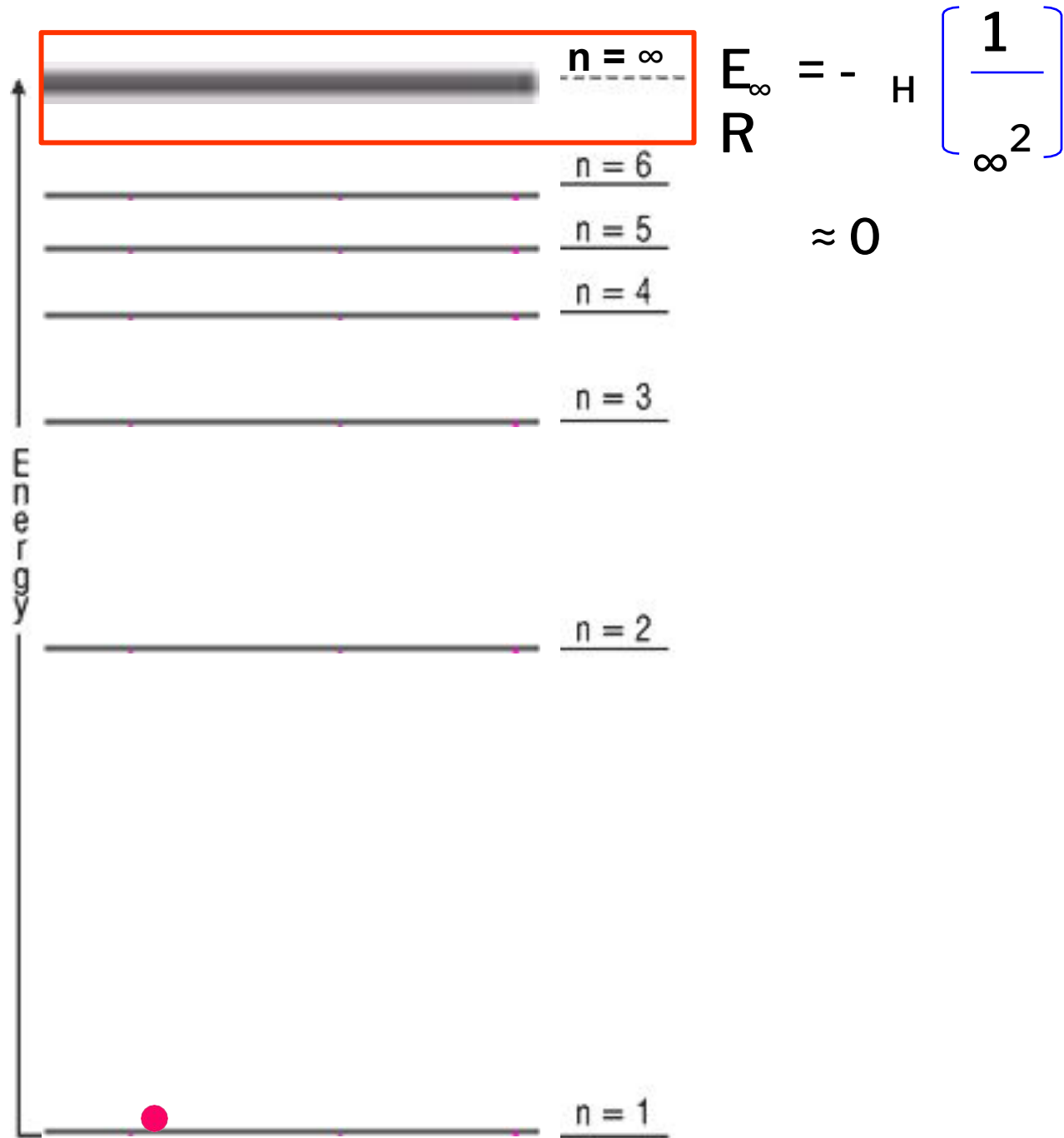
IONISATION ENERGY

Ionisation energy is the **minimum energy** required to remove **one mole of electron from one mole of gaseous atom or ion.**



THINGS TO NOTICE ABOUT THE EQUATION

- The state symbols - (g) - are essential. When you are talking about ionisation energies, everything must be present in the gas state.
- Ionisation energies are measured in kJ mol^{-1} (kilojoules per mole).
- The hydrogen atom is said to be ionised when **electron is removed from its ground state ($n = 1$) to $n = \infty$.**



IONISATION ENERGY

For Hydrogen Atom (Lyman series) :

- At $n = \infty$, the potential energy of electron is zero, here the **nucleus attractive force has no effect on the electron**; the electron is free from the nucleus force of attraction.
- The process of **removing electron** from **$n = 1$ to $n = \infty$** is called **ionisation process**.
- **Energy involved** during the process is called **ionisation energy**.

Example 12

What is meant by ionisation energy of hydrogen?

Suggested answer:

- **The minimum energy required to remove one mole of electron from one mole of gaseous atom to form one mole of gaseous positive ion.**

Example 13

Give a brief description on how the ionisation energy of hydrogen can be obtained from the line spectrum of hydrogen atom.

Suggested answer:

- **Ionisation energy is the energy that is required in removing the electron from a hydrogen atom from its ground state, $n = 1$ to $n = \infty$.**
- **When the electron were at $n = \infty$, there is no interactions between the electron and the nucleus and the energy is zero.**
- **Thus, the hydrogen atom is said to be ionised and becomes H^+ ion.**
- **Energy involved during the process is called ionisation energy.**

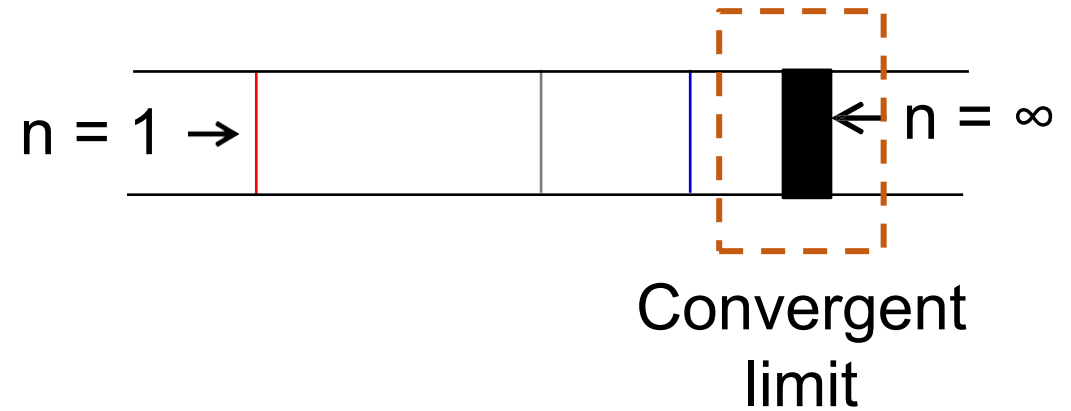
The ionisation energy can be
determined in two ways:

Rydberg's equation

$$\Delta E = R_H \left(\frac{1}{n_i^2} - \frac{1}{n_f^2} \right)$$

$$n_i = 1, n_f = \infty$$

Convergent limit



Example 14

Calculate the ionisation energy of hydrogen atom in **kJ mol⁻¹**.

Answer:

$$\begin{aligned}\Delta E_R &= H \left(\frac{1}{n_i^2} - \frac{1}{n_f^2} \right) \\ &= 2.18 \times 10^{-18} \left(\frac{1}{1^2} - \frac{1}{\infty^2} \right) \text{ J} \\ &= 2.18 \times 10^{-18} \text{ J per 1 electron}\end{aligned}$$

Remarks:

The electron **goes to infinity** i.e. it leaves the atom. Hence, we set $n_f = \infty$.

$$\text{IE} = (\text{energy for } 1\bar{e}) \times (\text{no. of } \bar{e} \text{ 1 mol})$$

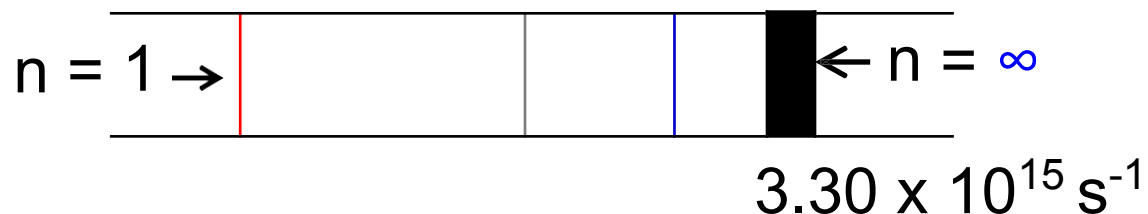
$$1 \text{ mol } \bar{e} \equiv 6.02 \times 10^{23} \bar{e}$$

$$\therefore \text{IE} = (6.02 \times 10^{23})(2.18 \times 10^{-18})$$

$$= 1312 \text{ kJ/mol}$$

Example 15

From the diagram below, calculate the ionisation energy for hydrogen using the converging frequency.



Answer:

When the electron is excited from $n=1$ to $n=\infty$, the converging frequency
 $= 3.30 \times 10^{15} \text{ s}^{-1}$

$$\Delta E = h\nu$$

$$= \left(\underset{\text{J}}{6.6256} \times 10^{-34} \right) \left(\underset{10}{3.30} \times 10^{15} \right)$$

$$= 2.18 \times 10^{-18} \text{ J per 1 electron}$$

$$\therefore \text{IE} = 2.186 \times 10^{-18} \times 6.02 \times 10^{23}$$

$$= 1316241 \text{ J/mol} = 1316$$

- Bohr's model was so successful that he immediately received world-wide fame.

-THE SUCCESS OF BOHR'S POSTULATES -

- Bohr's model worked for **hydrogen spectrum** or **any spectrum of ions** that contain **only one electron** such as ${}_2\text{He}^+$ or ${}_3\text{Li}^{2+}$.

-THE WEAKNESSES OF BOHR'S POSTULATES -

- His theory could not be extended to predict the energy levels and spectra of most atoms and ions having **more than one electron.**
- Electrons are **restricted to move in a certain distance** around the nucleus of an atom.
- His theory **cannot** explain the **formation of extra lines in the hydrogen spectrum.**
- His theory **cannot explain** the **dual nature of electron.**

THE DUAL NATURE OF ELECTRON

de Broglie's Postulate

In 1924 Louis de Broglie proposed particles such as electron has a **dual nature and possesses both wave and particle properties.**

de Broglie deduced that the particle is and wave properties are related by the expression:

$$\lambda = \frac{h}{mv}$$

Where: h = planck constant (Js)
 m = particle mass (kg)
 v = velocity (m/s)
 λ = wavelength

Heisenberg's uncertainty principle

It is **impossible** to know **simultaneously both momentum p** (defined as mass times velocity) and **position of a particle with certain.**

$$\Delta x \Delta p \geq \frac{h}{4\pi}$$

Δx = uncertainty in measuring the position

Δp = uncertainty in measuring the momentum

h = Planck constant

- When we apply this uncertainty principal to the hydrogen atom, we can see that in reality, **the electron does not orbit the nucleus in a well defined path** like we think it is. We actually cannot point where it is or how it moves exactly.
- We can **only calculate the probability of finding an electron** in a certain position at certain time.
- The **region in space** within which there is a **high probability of finding an electron** in an atom or molecule is known as an **orbital**.

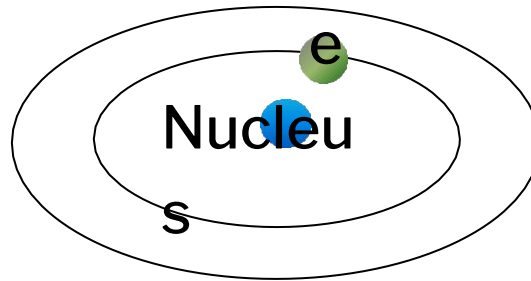


2.2

Quantum Mechanics

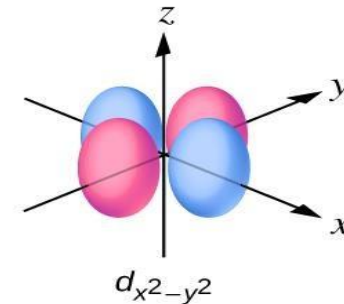
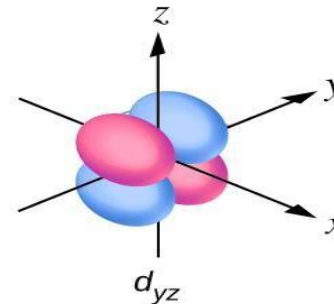
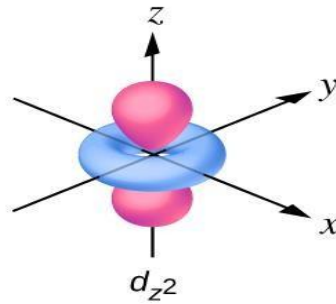
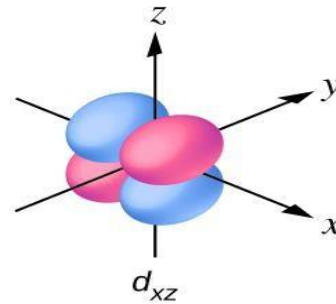
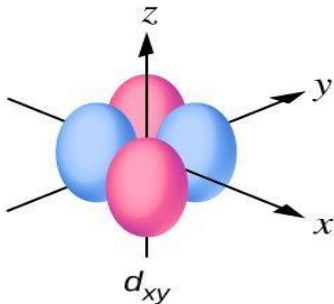
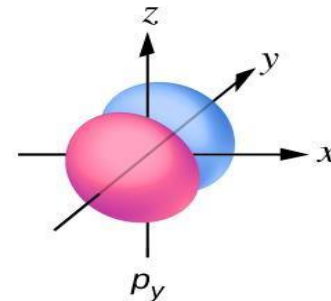
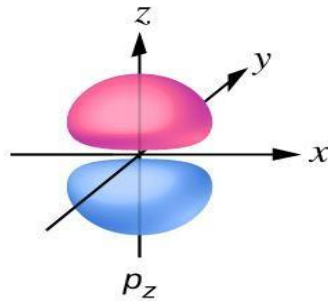
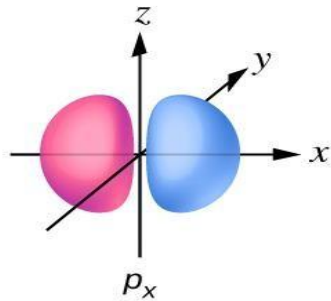
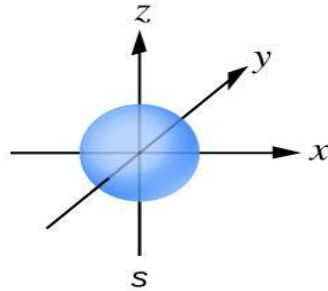
Orbit

The path of an electron around the nucleus of an atom.



DEFINITION OF ORBITAL

A region in space around the nucleus in which the probability of finding an electron is high.

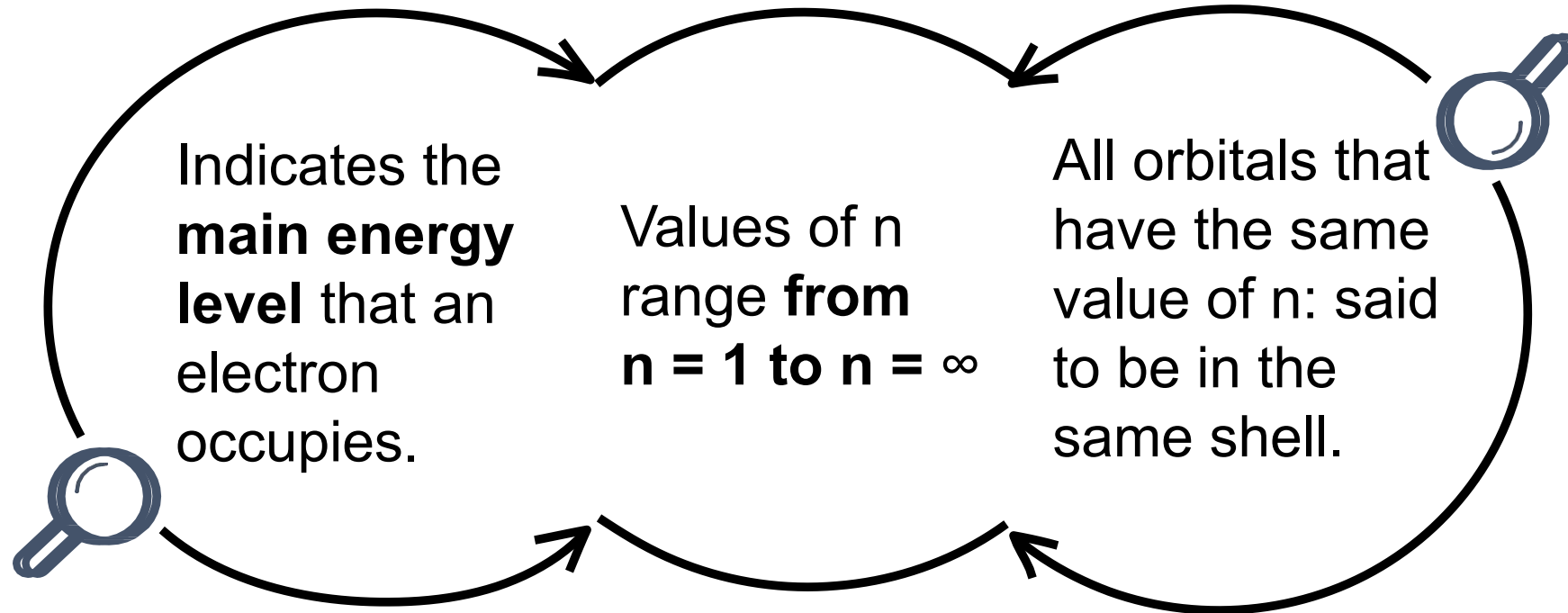


QUANTUM NUMBERS

Each electron in an atom is described and characterised by a set of **four quantum numbers**, namely as

- ✓ Principal quantum number (n) - size
- ✓ Angular momentum quantum number (l) - shape
- ✓ Magnetic quantum number (m) - orientation
- ✓ Spin quantum number (s) - spin

Principal Quantum Number, n



n is related to :

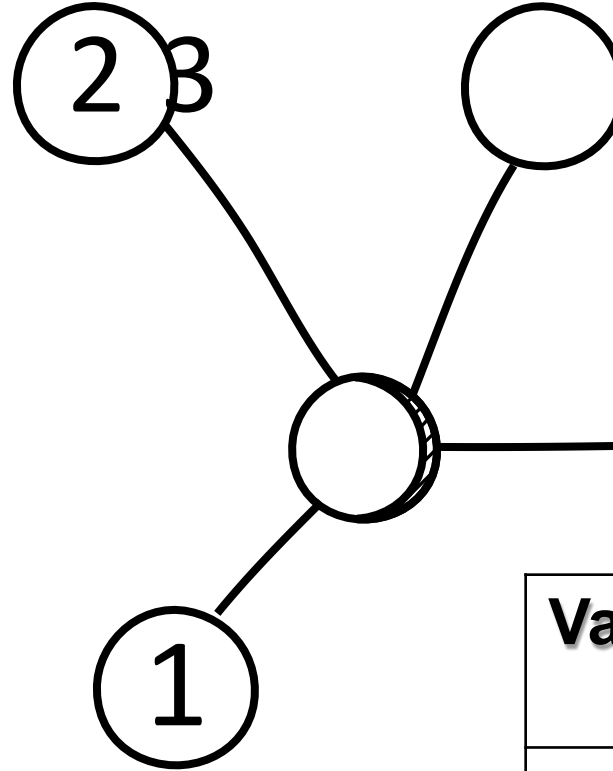
- the distance between the electron from the nucleus of the atom. ($n \uparrow$, average distance between electron and the nucleus $\uparrow$)
- the energy of the orbital. ($n \uparrow$, energies of the orbitals $\uparrow$)

When $n = 1$, it is called the first shell.

When $n = 2$, it is called the second shell.

Angular Momentum @ Azimuthal Quantum Number, l

For a given n , l can range from $l = 0$ to $l = (n - 1)$.



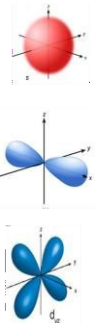
The value of l indicates the:

- **shape of the atomic orbital**
- **type of orbital**
- **angular momentum of the electron.**

For $l = 0$ to $l = 3$, a letter notation is given:

Divides the shells into smaller groups of orbitals: called **subshells**.

Value of l	Notation	Orbital shape
0	s	spherical
1	p	dumbbell
2	d	cloverleaf
3	f	

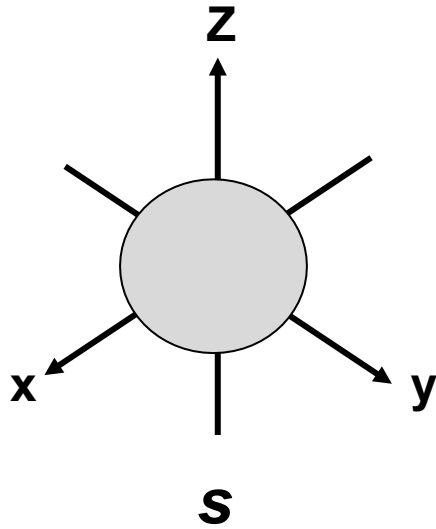


Angular Momentum @ Azimuthal Quantum Number, l

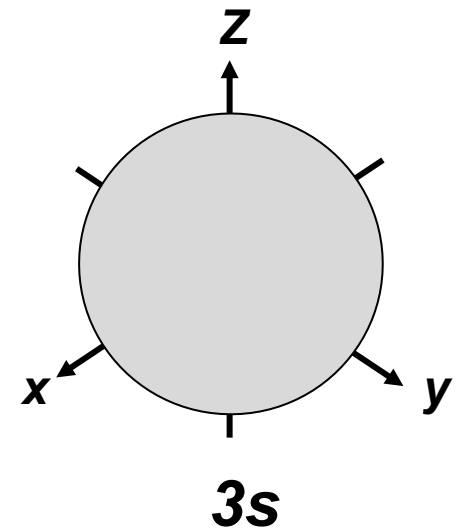
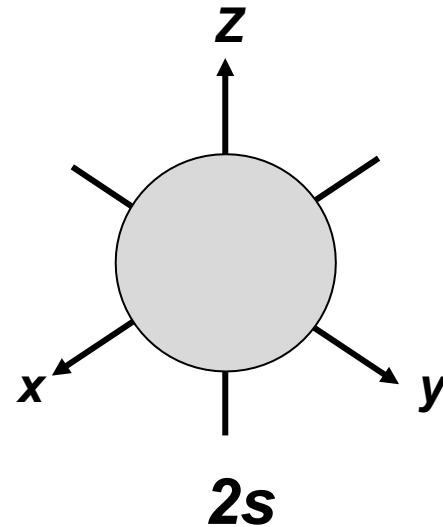
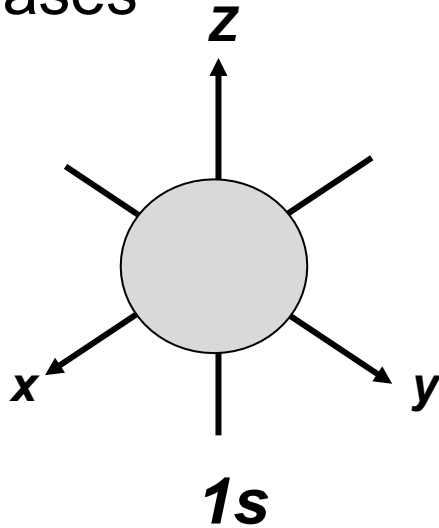
- Example:

n	l range from 0 to $(n - 1)$	Subshell
1	0	1s
2	0 1	2s 2p
3	0 1 2	3s 3p 3d

SHAPE OF "s" ORBITAL



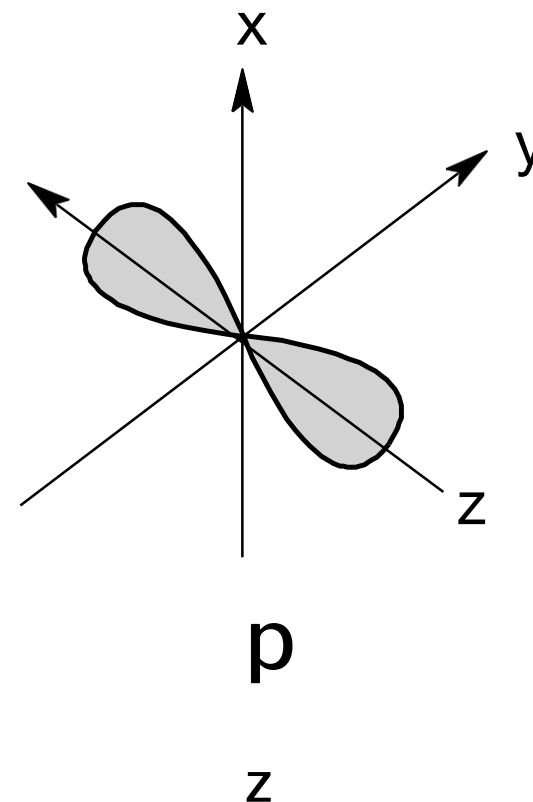
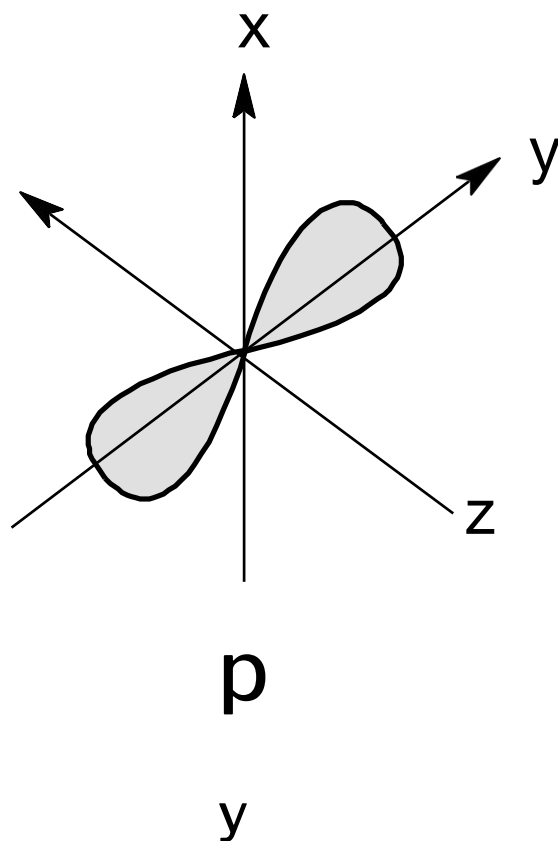
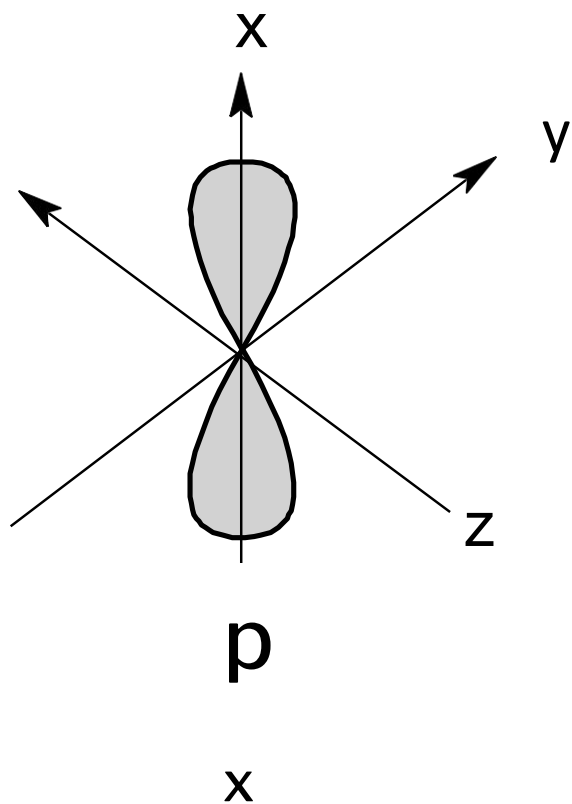
As value of n increases,
size of "s" orbital increases



SHAPE OF “p” ORBITAL

Designated as : p_x , p_y and p_z .

They have the **same shape** but **differ in their orientation in space**.



Example 1

a) For $n = 5$ what are the possible values of ℓ ?

Answer:

n	ℓ range from 0 to $(n - 1)$
5	$\ell = 0$ $\ell = 1$ $\ell = 2$ $\ell = 3$ $\ell = 4$

b) For $n = 3$ what are the possible values of ℓ and name each of these subshells?

Answer:

n	ℓ range from 0 to $(n - 1)$	Subshell
3	$\ell = 0$ $\ell = 1$ $\ell = 2$	3s 3p 3d

Magnetic Quantum Number, m

- 01 Describes the **orientation of orbital** in space.
- 02 The permitted value for m depends on the **value of ℓ** .
- 03 It has integer values ranging from $-\ell$ **through 0 to $+\ell$** .
- 04 An orbital can hold up to **two electrons**.
- 05 Divides the **subshells** into individual **orbitals**.

EXAMPLE:

$$\ell = 0$$

$$m = 0$$

$$\ell = 1$$

$$m = -1, 0, +1$$

$$\ell = 2$$

$$m = -2, -1, 0, +1, +2$$

$$\ell = 3$$

$$m = -3, -2, -1, 0, +1, +2, +3$$

Magnetic Quantum Number, m

Summary of Relationships Among The Quantum Numbers, n , l and m

Value of n	Value of l	Value of m	Subshell	No. of orbitals
1	0	0	1s	1
2	0	0	2s	1
	1	-1, 0, +1	2p	3
3	0	0	3s	1
	1	-1, 0, +1	3p	3
	2	-2, -1, 0, +1, +2	3d	5

Example 2

a) For $l = 3$ what are the possible values of m ?

$m = -3, -2, -1, 0, +1, +2, +3$

b) i) What is the designation for the subshell with $n = 6$ and $l = 1$

6p

ii) How many orbitals are there in this subshell?

There are three orbitals in this subshell

iii) Give the magnetic quantum number for each orbital?

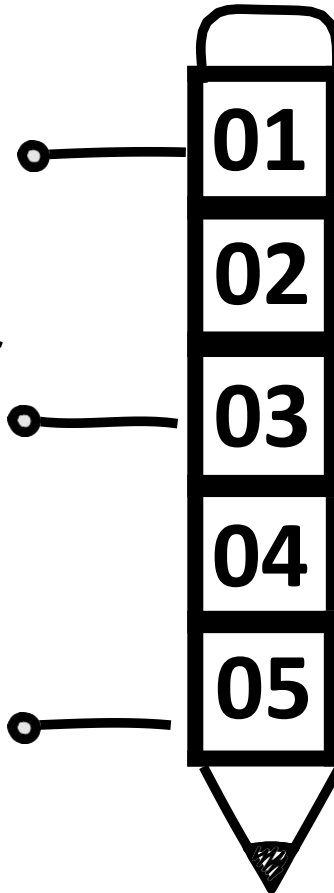
$m = -1, 0, +1$

Electron Spin Quantum Number, S

The maximum no. of electron that can be in a particular orbital is no more than **two electrons**.

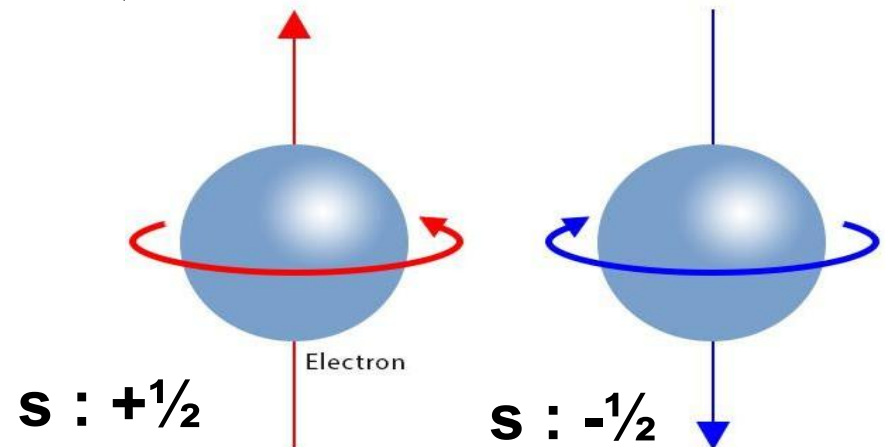
The spin quantum number represents the **spin of electron** on its own axis.

Electron behaves as if it were spinning.



Since the electron can only spin in either of two direction, s can have **only two values**.

2 possible values of **s** : $+\frac{1}{2}$ or $-\frac{1}{2}$, which can be clockwise or anticlockwise.



Summary of Relationships Among The Quantum Numbers n, l, m and s

n	l	m	s	Subshell	$\frac{2l+1}{2l+1}$ No. of orbitals	Max electrons
1	0	0	$\pm \frac{1}{2}$	1s	1	2
2	0	0	$\pm \frac{1}{2}$	2s	1	2
	1	-1, 0, +1	$\pm \frac{1}{2}$	2p	3	6
3	0	0	$\pm \frac{1}{2}$	3s	1	2
	1	-1, 0, +1	$\pm \frac{1}{2}$	3p	3	6
	2	-2, -1, 0, +1, +2	$\pm \frac{1}{2}$	3d	5	10
4	0	0	$\pm \frac{1}{2}$	4s	1	2
	1	-1, 0, +1	$\pm \frac{1}{2}$	4p	3	6
	2	-2, -1, 0, +1, +2	$\pm \frac{1}{2}$	4d	5	10
	3	-3, -2, -1, 0, +1, +2, +3	$\pm \frac{1}{2}$	4f	7	14

Example 3

a) What is the orbital of an electron having the following set of quantum numbers?

i) $n=2, l=1, m=0, s=-\frac{1}{2}$

ii) $n=1, l=0, m=0, s=+\frac{1}{2}$

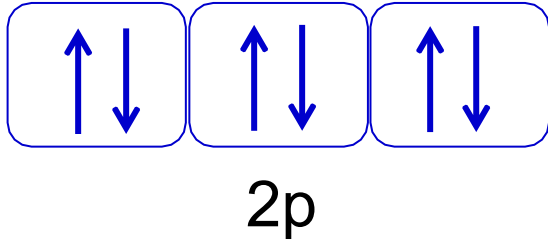
Answer:

i) 2p

ii) 1s

Example 4

Write all the possible combinations of quantum numbers for electron in a 2p orbital.



$\ell =$

1

$m = -1, 0, +1$

Answer:

$$n = 2, l = 1, m = -1, s = +\frac{1}{2}$$

$$n = 2, l = 1, m = -1, s = -\frac{1}{2}$$

$$n = 2, l = 1, m = 0, s = +\frac{1}{2}$$

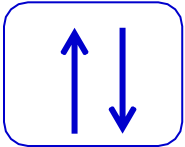
$$n = 2, l = 1, m = 0, s = -\frac{1}{2}$$

$$n = 2, l = 1, m = +1, s = +\frac{1}{2}$$

$$n = 2, l = 1, m = +1, s = -\frac{1}{2}$$

Example 5

Give a set of quantum numbers for an electron in **2s** orbital.



2s

$$\ell = 0$$

$$m = 0$$

Answer: $n = 2, l = 0, m = 0, s = +\frac{1}{2}$

or

$$n = 2, l = 0, m = 0, s = -\frac{1}{2}$$

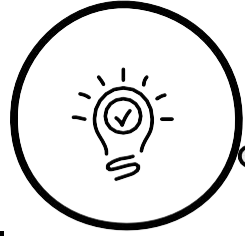


2.3

Electronic Configuration

ELECTRONIC CONFIGURATION

Shows **how electrons are filled** in the orbitals.

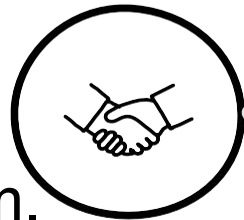


Describe the **distribution of electron** among the various orbital in an atom.



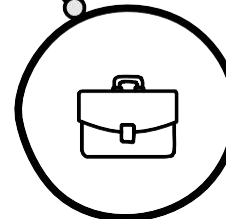
ELECTRON
IC
CONFIGURA
TION

Describe the **arrangement of electrons** in an atom.



Can be represented in 2 ways :

- **spdf notation**
- **orbital diagram**



s, p, d, f NOTATION

- **Uses numbers** to designate the principal shell.
- Letters **s, p, d or f** to **identify a subshell**.
- A **superscript** number following the letter **indicates the number of electron** in the designated subshell.

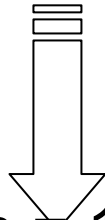
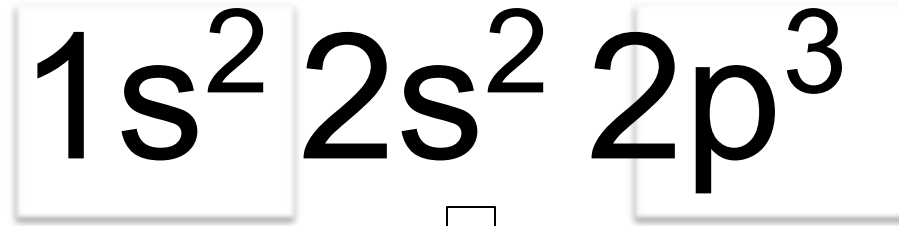
Principle quantum
number, n .

No. of electron in the
subshell.

1s²

Subshell @ angular momentum quantum
number, l .

Example 1



Principal quantum no., $n = 1$

Azimuthal quantum no., $l = 0$

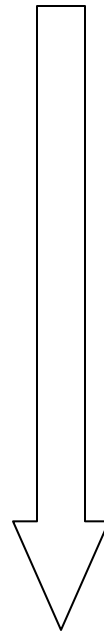
No. of electron in the subshell = 2



$n = 2$

$l = 1$

No. of electron in the subshell = 3



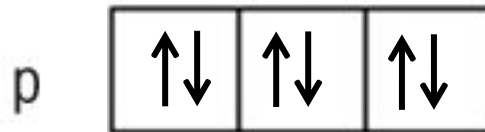
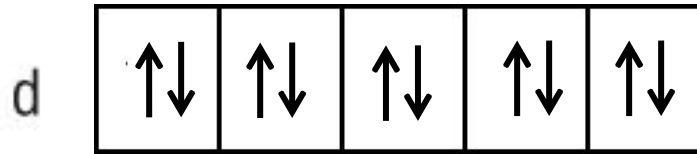
$n = 2$

$l = 0$

No. of electron in the subshell =
2

ORBITAL DIAGRAM

- Which **uses boxes/platforms to represent orbitals** within subshells.
- **Arrows** are used to represent electron .



There are 3 rules applied for assigning electrons to the orbital of an atom at the ground state.

01

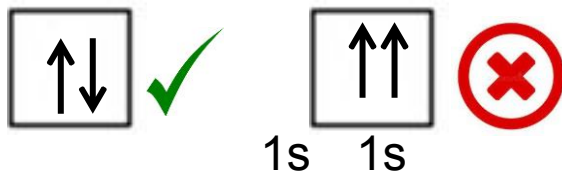
Electron will fill orbitals in the increasing order of orbital energy level.

The increasing order of energy are as $1s < 2s < 2p < 3s < 3p < 4s < 3d < 4p < 5s < 4d < 5p < 6s \dots$

02

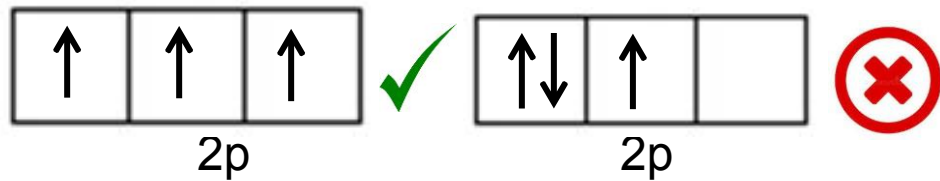
No two electrons in an atom can have the same set of four quantum numbers.

Only two electrons may occupy the same orbital, thus these electrons must have opposite spins



03

When electrons are filling to the orbital of equivalent energy, each orbital are filled singly with electron of the same spin first before pairing up.



Aufbau
Principle

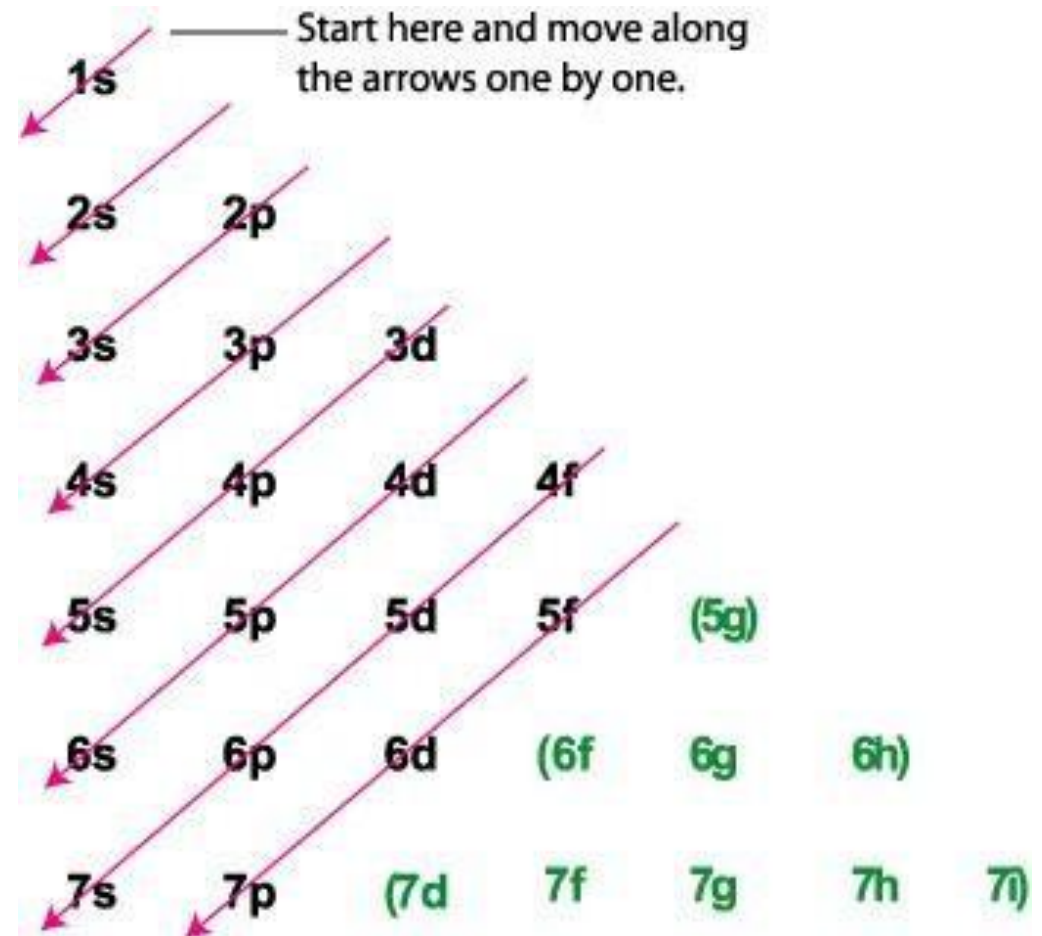
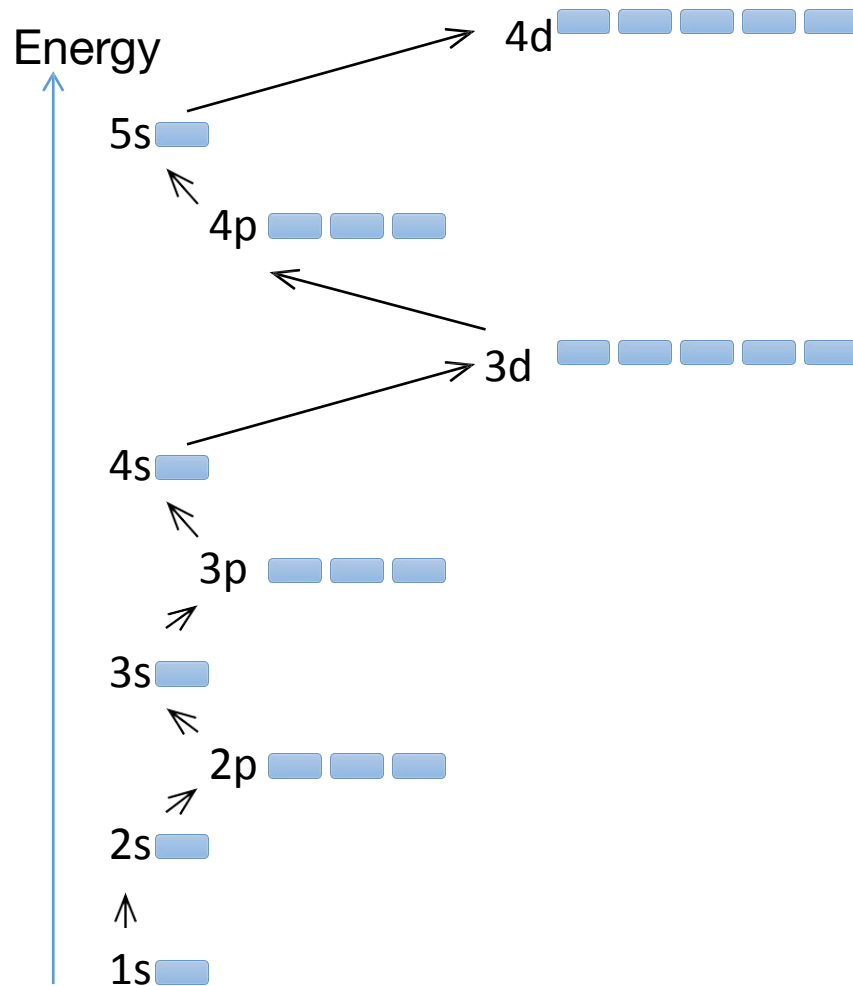
Pauli
Exclusion
Principle

Hund's
Rule

Aufbau Principle

The order of increasing energy is:

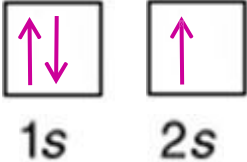
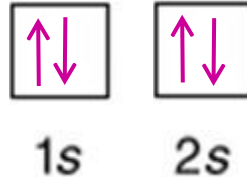
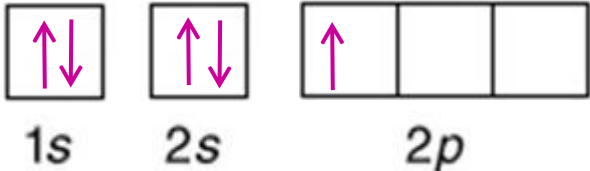
$1s < 2s < 2p < 3s < 3p < 4s < 3d < 4p < 5s < 4d < 5p < 6s \dots$



Example 2

Write the electronic configuration for the ${}_3\text{Li}$, ${}_4\text{Be}$ and ${}_5\text{B}$.

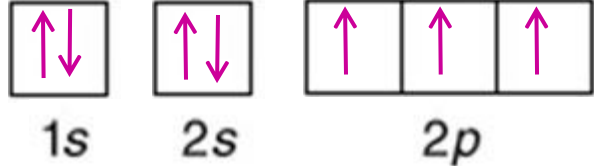
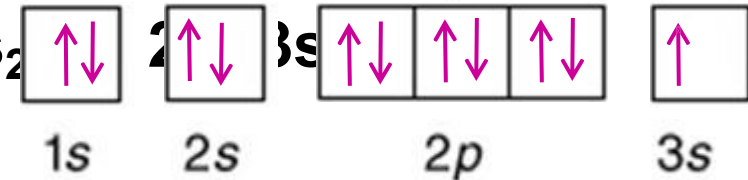
Answer:

Element	Total electron	Orbital diagram	Electronic configuration
${}_3\text{Li}$	3	 1s 2s	$1s^2 2s^1$
${}_4\text{Be}$	4	 1s 2s	$1s^2 2s^2$
${}_5\text{B}$	5	 1s 2s 2p	$1s^2 2s^2 2p^1$

Example 3

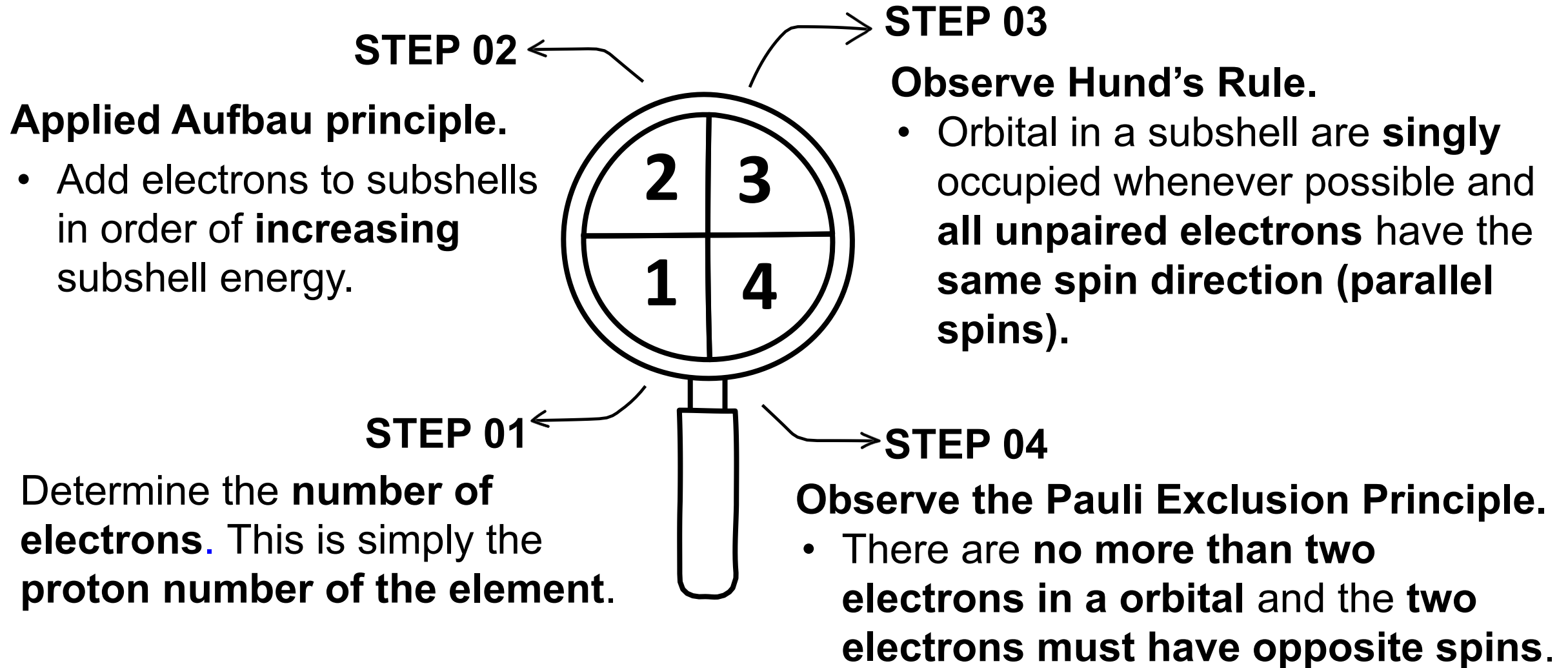
Write the electronic configuration for ${}_7\text{N}$ and ${}_{11}\text{Na}$.

Answer:

Element	Total electron	Orbital diagram	Electronic configuration
${}_7\text{N}$	7	 $1s$ $2s$ $2p$	$1s^2 2s^2 2p^3$
${}_{11}\text{Na}$	11	 $1s$ $2s$ $2p$ $3s$	

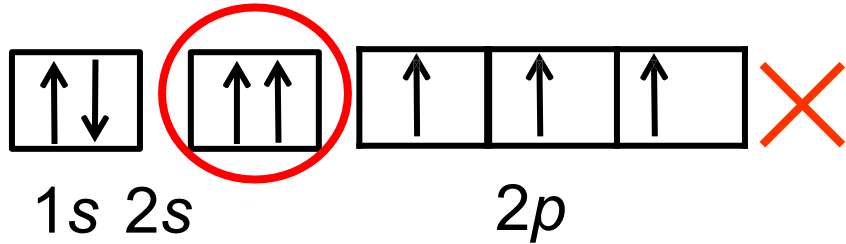
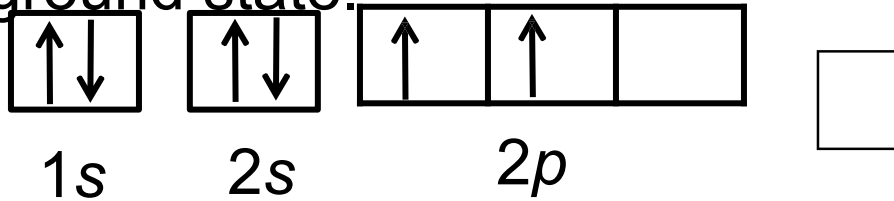
When electrons are added to the orbital of equivalent energy (degenerate orbital),
each orbital are filled singly with electron of the same spin first before it is pairing up
(Hund's Rule).

SUMMARY FOR WRITING ELECTRONIC CONFIGURATION FOR NEUTRAL ATOM

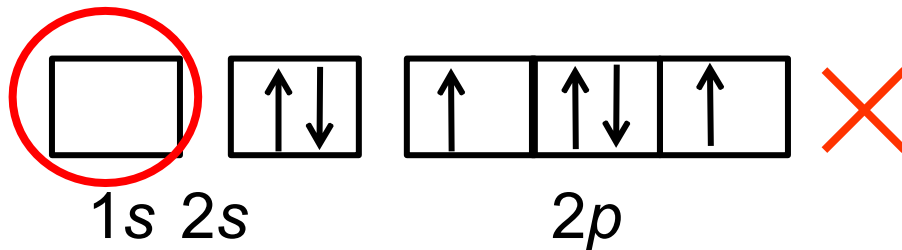


Example 4

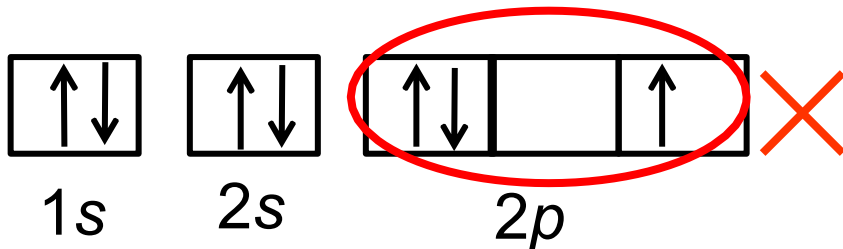
Determine whether the following orbital diagrams are acceptable for an atom in a ground state.



Does not obey Pauli Exclusion Principle



Does not obey Aufbau principle.



Does not obey Hund's Rule.

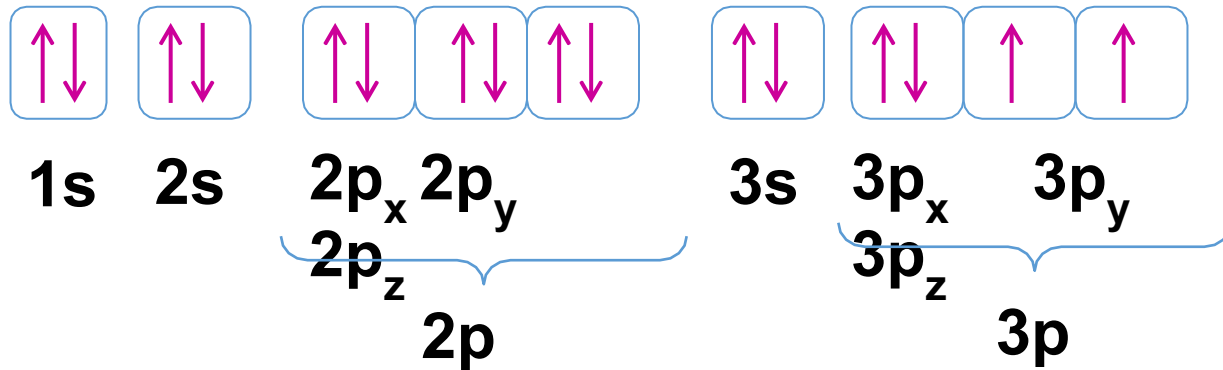
Example 5

State the electronic configurations for sulphur, using both the s, p, d, f notation and an orbital diagram. [Sulphur ($Z = 16$)]

Answer:

$Z = \text{number of electrons} = 16$

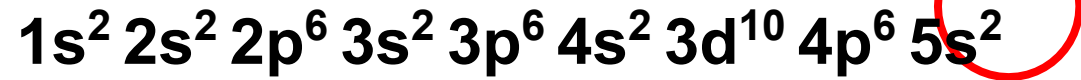
$1s^2 2s^2 2p^6 3s^2 3p^4$



Example 6

Write the electronic configuration of strontium, $_{38}\text{Sr}$ and give the values of the four quantum numbers for the electron(s) in its valence shell.

Answer:

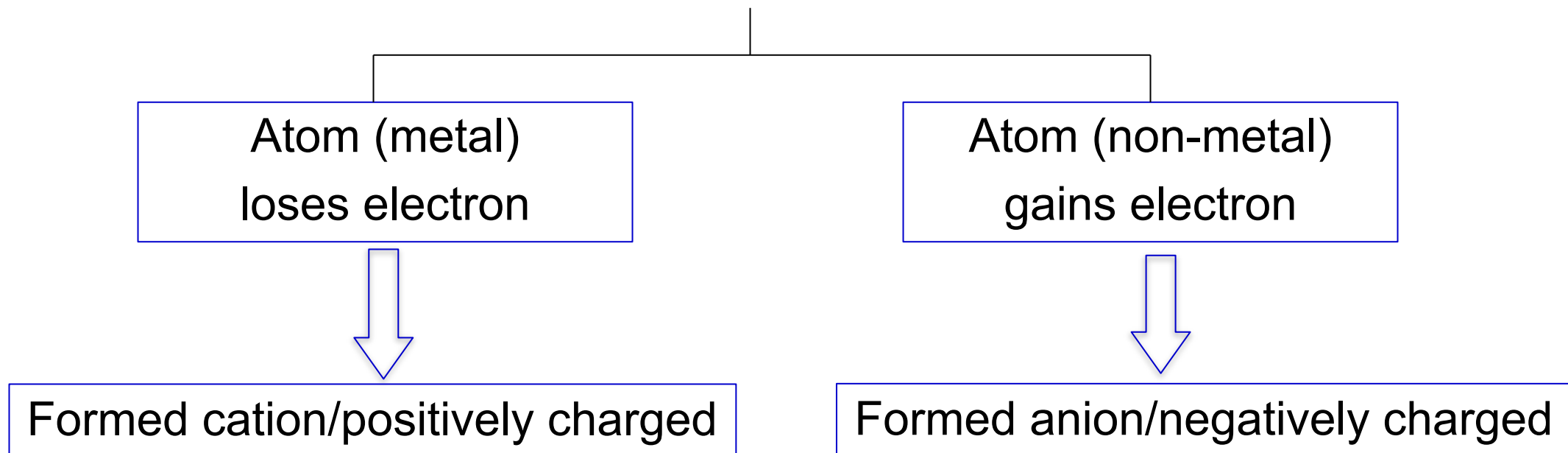


$$n = 5, l = 0, m = 0, s = +\frac{1}{2}$$

$$n = 5, l = 0, m = 0, s = -\frac{1}{2}$$

ELECTRONIC CONFIGURATION OF IONS

Ions are formed when :



Atom loses or gains electrons to form ion to achieve **octet** @ to achieve a **noble gas configuration which is very stable**.

Electrons are lost at or gained from the **outermost orbital**.

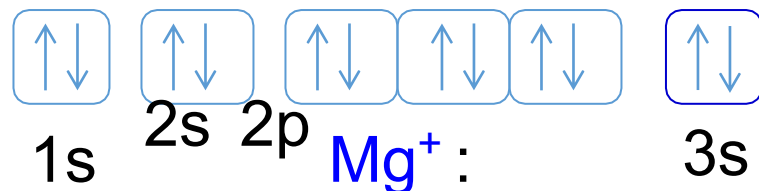
Example 7

Write the electronic configuration of the Mg and Mg^+ and Mg^{2+} .

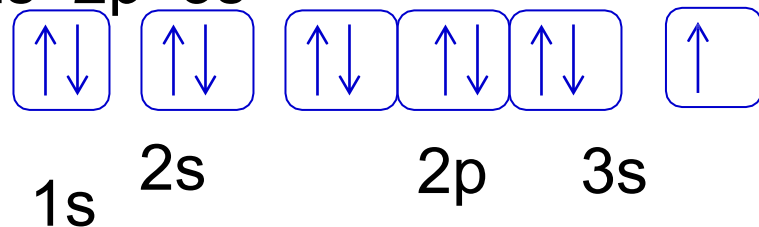
Answer:

Proton number = number of electrons =

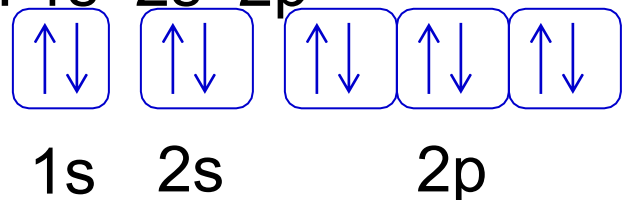
12 Mg : $1s^2 2s^2 2p^6 3s^2$



$1s^2 2s^2 2p^6 3s^1$



Mg^{2+} : $1s^2 2s^2 2p^6$ = electronic configuration of Ne



Mg is a metal



Formed ion by **losing electron.**



Electrons are **removed** from the **outermost orbital.**

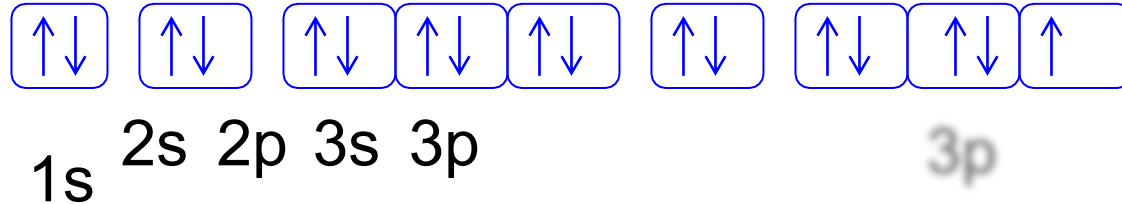
Example 8

Write the electronic configuration of the Cl and Cl⁻.

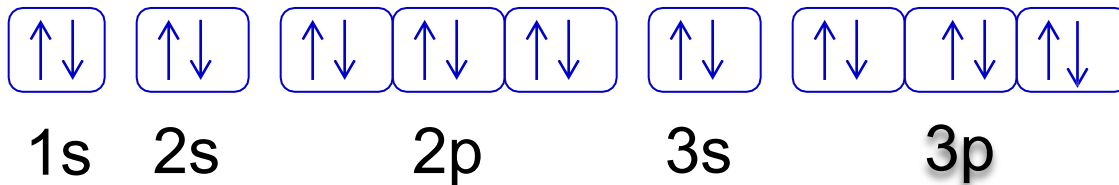
Answer:

Proton number = number of electrons =
17

${}_{17}^{17}\text{Cl}$: $1s^2 2s^2 2p^6 3s^2$
 $3p^5$



${}_{17}^{17}\text{Cl}^-$: $1s^2 2s^2 2p^6 3s^2 3p^6$ = electronic configuration of Ar



Cl is a non-metal



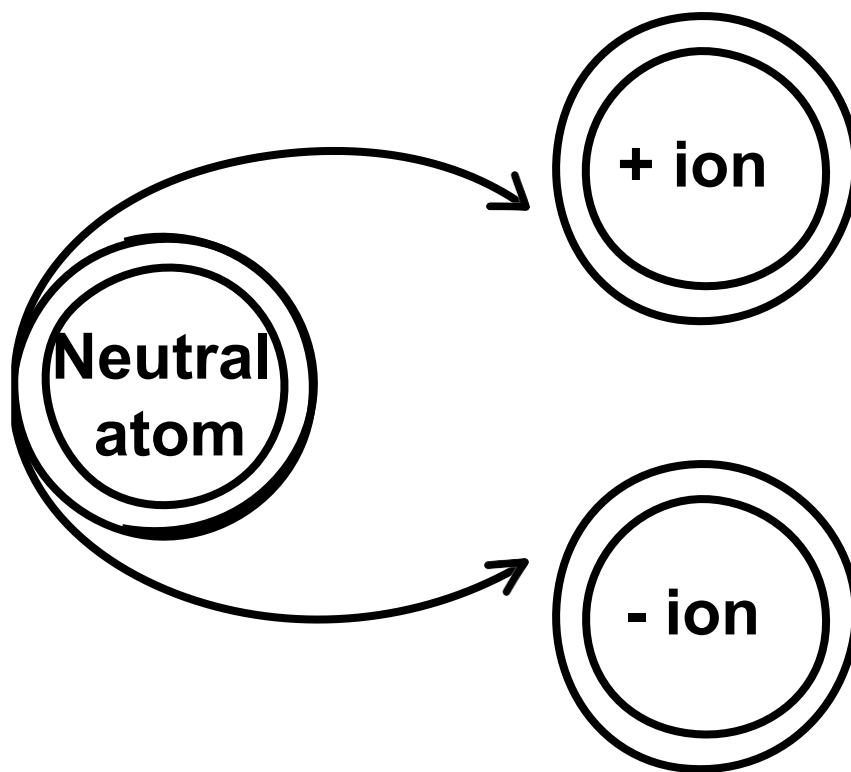
Formed ion by
accepting electron.



Electrons are
gained at the
outermost orbital.

SUMMARY FOR WRITING ELECTRONIC CONFIGURATION FOR ION

Start with the electronic structure of the neutral atom



Subtract
electrons for a
positive ion

Add electrons
for a negative
ion

THE ELECTRONIC CONFIGURATIONS OF THE TRANSITION ELEMENTS

The transition elements are found in the fourth row of the Periodic Table.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18												
1	1 H Hydrogen 1.00794																	2 He Helium 4.002602												
2	3 Li Lithium 6.941	4 Be Beryllium 9.012182											5 B Boron 10.811	6 C Carbon 12.011	7 N Nitrogen 14.0067	8 O Oxygen 15.9994	9 F Fluorine 18.9984632	10 Ne Neon 20.1797												
3	11 Na Sodium 22.98976928	12 Mg Magnesium 24.3050											13 Al Aluminum 26.9815386	14 Si Silicon 28.0855	15 P Phosphorus 30.973762	16 S Sulfur 32.065	17 Cl Chlorine 35.453	18 Ar Argon 39.948												
4	19 K Potassium 39.0983	20 Ca Calcium 40.078	21 Sc Scandium 44.955912	22 Ti Titanium 47.887	23 V Vanadium 50.9415	24 Cr Chromium 51.9961	25 Mn Manganese 54.938045	26 Fe Iron 55.845	27 Co Cobalt 58.933195	28 Ni Nickel 58.6934	29 Cu Copper 63.546	30 Zn Zinc 65.38	31 Ga Gallium 69.723	32 Ge Germanium 72.64	33 As Arsenic 74.92160	34 Se Selenium 78.96	35 Br Bromine 79.904	36 Kr Krypton 83.796												
5	37 Rb Rubidium 85.4678	38 Sr Strontium 87.62	39 Y Yttrium 88.90585	40 Zr Zirconium 91.224	41 Nb Niobium 92.90638	42 Mo Molybdenum 95.96	43 Tc Technetium (97.9072)	44 Ru Ruthenium 101.07	45 Rh Rhodium 102.90550	46 Pd Palladium 106.42	47 Ag Silver 107.8682	48 Cd Cadmium 112.411	49 In Indium 114.818	50 Sn Tin 118.710	51 Sb Antimony 121.760	52 Te Tellurium 127.60	53 I Iodine 126.90447	54 Xe Xenon 131.29												
6	55 Cs Cesium 132.9054519	56 Ba Barium 137.327	57–71	72 Hf Hafnium 178.49	73 Ta Tantalum 180.94788	74 W Tungsten 183.84	75 Re Rhenium 186.207	76 Os Osmium 190.23	77 Ir Iridium 192.225	78 Pt Platinum 195.084	79 Au Gold 196.966569	80 Hg Mercury 200.59	81 Tl Thallium 204.3833	82 Pb Lead 207.2	83 Bi Bismuth 208.98040	84 Po Polonium (209)	85 At Astatine (210)	86 Rn Radon (222)												
7	87 Fr Francium (223)	88 Ra Radium (226)	89–103	104 Rf Rutherfordium (261)	105 Db Dubnium (262)	106 Sg Seaborgium (266)	107 Bh Bohrium (264)	108 Hs Hassium (277)	109 Mt Meitnerium (268)	110 Ds Darmstadtium (271)	111 Rg Roentgenium (272)	112 Uub Ununbium (285)	113 Uut Ununtrium (284)	114 Uuq Ununquadium (289)	115 Uup Ununpentium (288)	116 Uuh Ununhexium (292)	117 Uus Ununseptium (293)	118 Uuo Ununoctium (294)												
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	57 La Lanthanum 138.90547	58 Ce Cerium 140.116	59 Pr Praseodymium 140.90765	60 Nd Neodymium 144.242	61 Pm Promethium (145)	62 Sm Samarium 150.36	63 Eu Europium 151.964	64 Gd Gadolinium 157.25	65 Tb Terbium 158.92535	66 Dy Dysprosium 162.500	67 Ho Holmium 164.93032	68 Er Erbium 167.259	69 Tm Thulium 168.93421	70 Yb Ytterbium 173.054	71 Lu Lutetium 174.9668	89 Ac Actinium (227)	90 Th Thorium 232.03806	91 Pa Protactinium 231.03688	92 U Uranium 238.02891	93 Np Neptunium (237)	94 Pu Plutonium (244)	95 Am Americium (243)	96 Cm Curium (247)	97 Bk Berkelium (247)	98 Cf Californium (251)	99 Es Einsteinium (252)	100 Fm Fermium (257)	101 Md Mendelevium (258)	102 No Nobelium (259)	103 Lr Lawrencium (262)

Electronic Configurations of the First Row of Transition Elements

Element	1s	2s	2p	3s	3p	4s	3d	
²¹ Sc	↑↓	↑↓	↑↓ ↑↓ ↑↓	↑↓	↑↓	↑↓	↑↓ ↑	
²² Ti	↑↓	↑↓	↑↓ ↑↓ ↑↓	↑↓	↑↓ ↑↓ ↑↓	↑↓	↑ ↑	
²³ V	↑↓	↑↓	↑↓ ↑↓ ↑↓	↑↓	↑↓ ↑↓ ↑↓	↑↓	↑ ↑ ↑	
²⁴ Cr	↑↓	↑↓	↑↓ ↑↓ ↑↓	↑↓	↑↓ ↑↓ ↑↓	↑	↑ ↑ ↑ ↑ ↑	
²⁵ Mn	↑↓	↑↓	↑↓ ↑↓ ↑↓	↑↓	↑↓ ↑↓ ↑↓	↑↓	↑ ↑ ↑ ↑ ↑	
²⁶ Fe	↑↓	↑↓	↑↓ ↑↓ ↑↓	↑↓	↑↓ ↑↓ ↑↓	↑↓	↑↓ ↑ ↑ ↑ ↑	
²⁷ Co	↑↓	↑↓	↑↓ ↑↓ ↑↓	↑↓	↑↓ ↑↓ ↑↓	↑↓	↑↓ ↑↓ ↑↓ ↑ ↑ ↑	
²⁸ Ni	↑↓	↑↓	↑↓ ↑↓ ↑↓	↑↓	↑↓ ↑↓ ↑↓	↑↓	↑↓ ↑↓ ↑↓ ↑↓ ↑ ↑	
²⁹ Cu	↑↓	↑↓	↑↓ ↑↓ ↑↓	↑↓	↑↓ ↑↓ ↑↓	↑	↑↓ ↑↓ ↑↓ ↑↓ ↑↓ ↑↓	
³⁰ Zn	↑↓	↑↓	↑↓ ↑↓ ↑↓	↑↓	↑↓ ↑↓ ↑↓	↑↓	↑↓ ↑↓ ↑↓ ↑↓ ↑↓ ↑↓	

THE ANOMALOUS ELECTRONIC CONFIGURATIONS

The 3 rules applied for assigning electron to the orbital :



1. **The Aufbau Principle**
2. **The Pauli Exclusion Principle**
3. **The Hund's Rule**



However, there are some atoms or ions that **do not obey** the 3 rules.



The electronic configuration that **do not obey** the 3 rules is known as the **anomalous electronic configuration**.

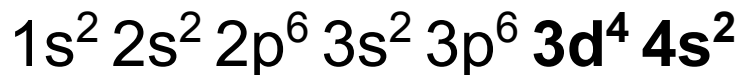


These out of ordinary electronic configurations exist for atoms or ions with the **3d orbitals**.

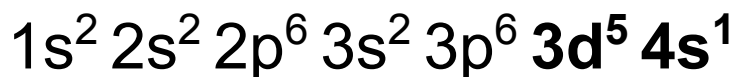
THE ANOMALOUS ELECTRONIC CONFIGURATIONS

1 Chromium: $_{24}\text{Cr}$

▪ Predicted configuration

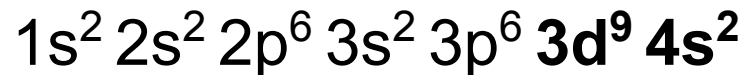


▪ Actual configuration

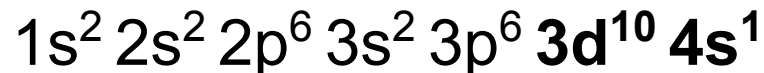


2 Copper : $_{29}\text{Cu}$

▪ Predicted configuration



▪ Actual configuration



EXPLANATION:

The energy of 4s and 3d are very close. Thus, one electron from 4s orbital is promoted to 3d orbital.

Hence, it has an **extra stability** of a **half-filled d orbital** ($3d^5$)

The energy of 4s and 3d are very close. Thus, one electron from 4s orbital is promoted to 3d orbital.

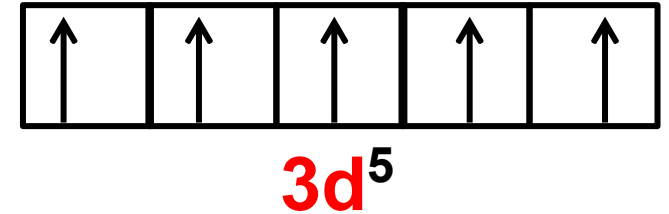
Hence, it has an **extra stability** of a **completely-filled d orbital** ($3d^{10}$)

A completely-filled and half filled orbital has extra stability because

Half filled orbital:

Is referred to the condition where all of the $3d$ orbitals are filled with one single electron.

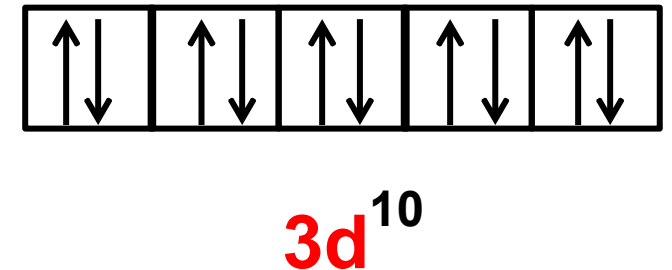
In half-filled orbital, again, the **electrons are evenly distributed and repulsion between electrons are minimised.**



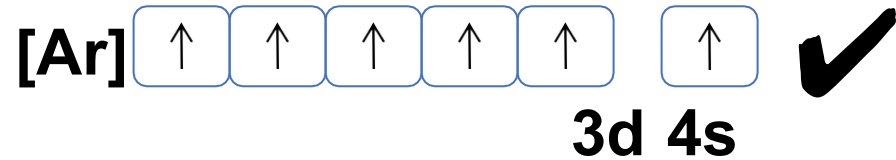
Completely filled orbital:

Is referred to the condition where all of the $3d$ orbitals are fully filled with electrons.

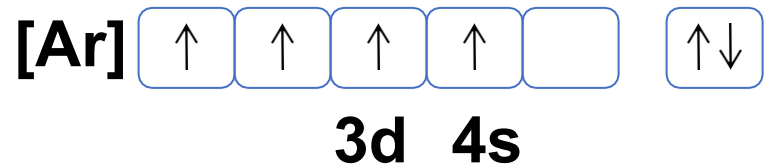
In completely-filled orbital, the **electrons are spread evenly around the nucleus.** This makes an atom stable.



Chromium, Cr

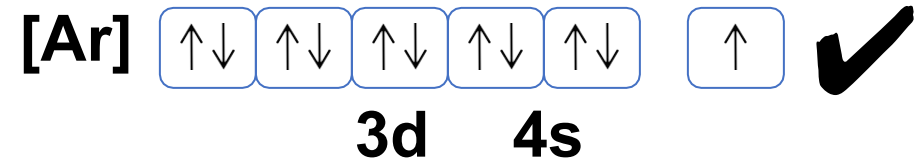


has extra stability than

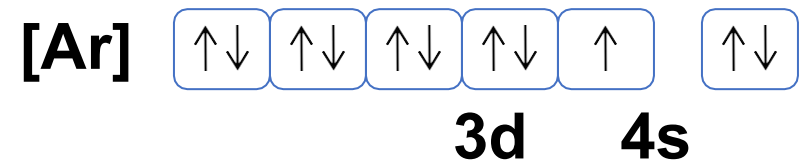


because of the
Half filled “d” orbital.

Copper, Cu



has extra stability than



because of the
Completely filled “d” orbital.

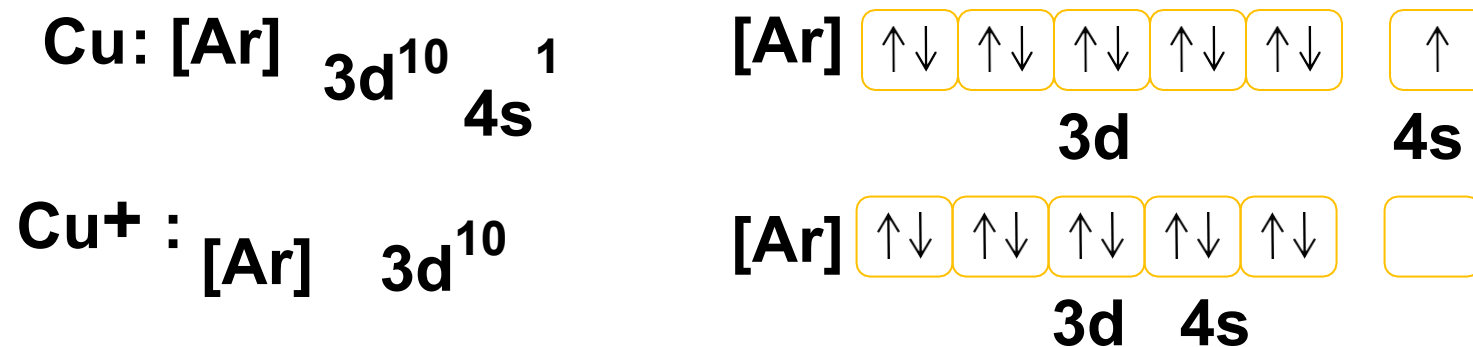
- Transition elements are all metals.
- They loss electrons to form ions.
- Electrons are lost from the outermost orbital, i.e. the 4s orbital.

Example 9

Write the valence electronic configuration of the Cu and Cu⁺.

Answer:

Proton number = number of electrons = 29



Electrons are removed from the **outermost orbital**.

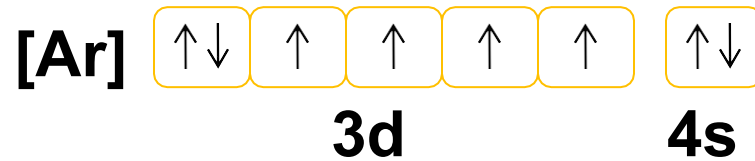
Example 10

Write the valence electronic configuration of the Fe, Fe²⁺ and Fe³⁺.

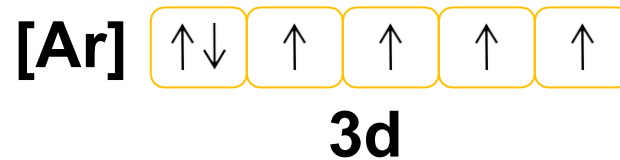
Answer:

Proton number = number of electrons = 26

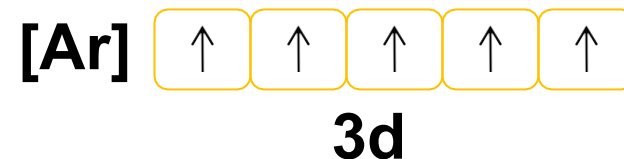
Fe : [Ar] 3d
6 4s²



Fe₂₊ : [Ar] 3d
6



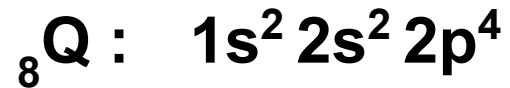
Fe³⁺ : [Ar] 3d
5



Example 11

An element Q has proton number of 8. Write the electronic configuration of the element. Explain two rules applied in arranging the electrons in the orbitals.

Answer:



Aufbau Principle

Electron will fill orbitals in the increasing order of orbital energy level.

The increasing order of energy are as $1s < 2s < 2p < 3s < 3p < 4s < 3d < 4p < 5s < 4d < 5p < 6s \dots$

Pauli Exclusion Principle

No two electrons in an atom can have the same set of four quantum numbers. Only two electrons may occupy the same orbital, thus these electrons must have opposite spins

Hund's Rule

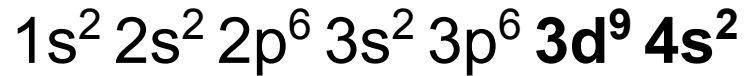
When electrons are filling to the orbital of equivalent energy, each orbital are filled singly with electron of the same spin first before pairing up.

Example 12

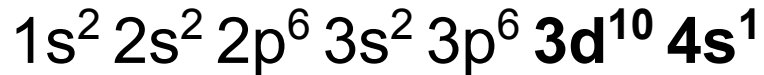
The proton number of copper is 29. Write the valence electronic configuration of the copper atom. Give a set of quantum numbers for the valence electron that occupy the s orbital of copper.

Answer:

- Predicted electronic configuration of Cu



- Actual electronic configuration of Cu



- Valence electronic configuration of Cu



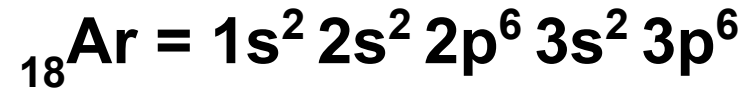
- A set of quantum numbers for the valence electron that occupy the s orbital of copper:

$$\mathbf{n = 4, l = 0, m = 0, s = +\frac{1}{2} \quad \text{or} \quad n = 4, l = 0, m = 0, s = -\frac{1}{2}}$$

Example 13

Write an electronic configuration for argon and give two oppositely charged ions having the same electronic configuration as argon.

Answer:



or





THE END